

Perioperative infection prediction -- subgroup deployment-readiness audit (audited cohort (2014–2022))

Model	Random forest (M2, end of surgery), published out-of-fold predictions
Cohort	Inselspital Bern -- audited cohort (2014–2022): n=10,719; 504 events
Outcome	Present-on-Admission-coded postoperative bacterial infection
Validation	5-fold stratified CV with SMOTE inside training folds (OOF predictions)
Threshold	0.1 (reference operating anchor, not an elicited optimum)
Uncertainty	2,000 patient-level bootstrap resamples; pre-registered event floors applied (AUROC 25, calibration intercept 25, calibration slope 100, net benefit 50)
Mitigation	Not shown here. The mitigation analysis is the cross-fitted recalibration ladder reported in Table 4 and Supplementary Figure 1; a split-based before/after would be a different analysis.
Window	all years (2014-2022)

Audit summary

N: 10719	Events: 504	Prevalence: 0.047	Threshold: 0.100
Bootstrap resamples: 2000	Resampling: subgroup-stratified patient-level	Attributes: 9	

Cohort characterisation (Table 1)

6 of 8 variables differ significantly across 'sex' at FDR $q < 0.05$ (Benjamini–Hochberg).

		Grouped by sex			P-Value	q_fdr
		Overall	Female	Male		
n		10719	3689	7030		
age, median [Q1,Q3]		66.0 [55.0,74.0]	66.0 [53.0,75.0]	66.0 [56.0,73.0]	0.467	0.467
age_group, n (%)	0-17	367 (3.4)	175 (4.7)	192 (2.7)	<0.001	0.0016
	18-44	941 (8.8)	408 (11.1)	533 (7.6)		
	45-64	3715 (34.7)	1179 (32.0)	2536 (36.1)		
	65-79	4684 (43.7)	1470 (39.8)	3214 (45.7)		
	80+	1012 (9.4)	457 (12.4)	555 (7.9)		
asa, n (%)	1	1227 (11.4)	630 (17.1)	597 (8.5)	<0.001	0.0016
	2	213 (2.0)	78 (2.1)	135 (1.9)		
	3	3803 (35.5)	1549 (42.0)	2254 (32.1)		
	4	5476 (51.1)	1432 (38.8)	4044 (57.5)		
frailty, n (%)	frail	6139 (57.3)	1666 (45.2)	4473 (63.6)	<0.001	0.0016
	non_frail	1022 (9.5)	522 (14.2)	500 (7.1)		
	pre_frail	3558 (33.2)	1501 (40.7)	2057 (29.3)		
emergencysurgery, n (%)	0	9250 (86.3)	3221 (87.3)	6029 (85.8)	0.028	0.0373
	1	1469 (13.7)	468 (12.7)	1001 (14.2)		
duration_of_surg, median [Q1,Q3]		213.0 [147.0,281.0]	193.0 [135.0,267.0]	221.0 [156.0,285.0]	<0.001	0.0016
surgical_group, n (%)	chopcluster_1	903 (8.4)	365 (9.9)	538 (7.7)	<0.001	0.0016
	chopcluster_10	346 (3.2)	98 (2.7)	248 (3.5)		
	chopcluster_11	303 (2.8)	303 (8.2)	0 (0.0)		
	chopcluster_12	1623 (15.1)	711 (19.3)	912 (13.0)		
	chopcluster_13	213 (2.0)	95 (2.6)	118 (1.7)		
	chopcluster_3	807 (7.5)	317 (8.6)	490 (7.0)		
	chopcluster_4	3764 (35.1)	836 (22.7)	2928 (41.7)		
	chopcluster_5	769 (7.2)	189 (5.1)	580 (8.3)		
	chopcluster_6	530 (4.9)	249 (6.7)	281 (4.0)		

Grouped by sex						
		Overall	Female	Male	P-Value	q_fdr
	chopcluster_7	298 (2.8)	105 (2.8)	193 (2.7)		
	chopcluster_8	306 (2.9)	95 (2.6)	211 (3.0)		
	none	857 (8.0)	326 (8.8)	531 (7.6)		
outcome, n (%)	0	10215 (95.3)	3499 (94.8)	6716 (95.5)	0.123	0.141
	1	504 (4.7)	190 (5.2)	314 (4.5)		

Per-variable effect sizes and BH-FDR

Variable	Kind	n	p-value	q (FDR)	Effect size	Kind	Magnitude
duration_of_surg	continuous	10719	0.001	0.0016	-0.161	d	negligible
frailty	categorical	10719	0.001	0.0016	+0.185	v	small
age_group	categorical	10719	0.001	0.0016	+0.116	v	small
asa	categorical	10719	0.001	0.0016	+0.191	v	small
surgical_group	categorical	10719	0.001	0.0016	+0.306	v	medium
emergencysurgery	binary	10719	0.028	0.0373	-0.046	h	negligible
outcome	binary	10719	0.123	0.1406	+0.032	h	negligible
age	continuous	10719	0.467	0.4670	-0.079	d	negligible

Discrimination

Attribute	Subgroup	n	Events	Prevalence	AUROC (95% CI)	AUPRC (95% CI)	Sensitivity (95% CI)	Specificity (95% CI)
sex	Overall	10719	504	0.047	0.876 (0.859–0.892)	0.376 (0.332–0.422)	0.899 (0.871–0.925)	0.647 (0.638–0.656)
sex	female	3689	190	0.052	0.878 (0.849–0.903)	0.408 (0.341–0.478)	0.900 (0.856–0.940)	0.640 (0.624–0.656)
sex	male	7030	314	0.045	0.875 (0.854–0.895)	0.356 (0.303–0.411)	0.898 (0.863–0.930)	0.650 (0.639–0.662)
age_group	Overall	10719	504	0.047	0.876 (0.859–0.892)	0.376 (0.332–0.422)	0.899 (0.871–0.925)	0.647 (0.638–0.656)
age_group	0-17	367	21	0.057	0.807	0.246	0.857	0.462
age_group	18-44	941	34	0.036	0.854 (0.792–0.908)	0.329 (0.175–0.505)	0.824 (0.679–0.939)	0.682 (0.653–0.712)
age_group	45-64	3715	147	0.040	0.888 (0.856–0.916)	0.382 (0.303–0.468)	0.918 (0.872–0.959)	0.666 (0.650–0.682)
age_group	65-79	4684	249	0.053	0.889 (0.865–0.910)	0.438 (0.371–0.509)	0.912 (0.875–0.944)	0.654 (0.639–0.668)
age_group	80+	1012	53	0.052	0.805 (0.744–0.863)	0.227 (0.145–0.344)	0.849 (0.746–0.940)	0.577 (0.545–0.607)
frailty	Overall	10719	504	0.047	0.876 (0.859–0.892)	0.376 (0.332–0.422)	0.899 (0.871–0.925)	0.647 (0.638–0.656)
frailty	frail	6139	400	0.065	0.878 (0.860–0.896)	0.428 (0.379–0.481)	0.927 (0.901–0.951)	0.584 (0.571–0.597)
frailty	non_frail	1022	6	0.006	0.868	0.084	0.833	0.816
frailty	pre_frail	3558	98	0.028	0.821 (0.774–0.865)	0.190 (0.127–0.274)	0.786 (0.701–0.863)	0.702 (0.687–0.716)
clinic	Overall	10719	504	0.047	0.876 (0.859–0.892)	0.376 (0.332–0.422)	0.899 (0.871–0.925)	0.647 (0.638–0.656)
clinic	ORL	233	15	0.064	0.820	0.221	1.000	0.408
clinic	gynecology	452	24	0.053	0.821	0.395	0.750	0.689
clinic	neuro	1117	26	0.023	0.871 (0.785–0.939)	0.434 (0.240–0.621)	0.808 (0.643–0.955)	0.756 (0.731–0.780)
clinic	ortho	1690	76	0.045	0.852 (0.808–0.887)	0.232 (0.160–0.332)	0.908 (0.835–0.965)	0.633 (0.610–0.657)
clinic	pediatric	65	3	0.046	0.653	0.181	0.667	0.452
clinic	thoracic	723	14	0.019	0.945	0.411	0.929	0.781
clinic	urology	349	19	0.054	0.785	0.331	0.737	0.624
clinic	vascular	4852	184	0.038	0.866 (0.834–0.896)	0.323 (0.257–0.395)	0.886 (0.837–0.929)	0.656 (0.643–0.670)
clinic	visceral	1238	143	0.116	0.883 (0.854–0.909)	0.557 (0.472–0.640)	0.965 (0.932–0.993)	0.482 (0.453–0.510)
surgical_group	Overall	10719	504	0.047	0.876 (0.859–0.892)	0.376 (0.332–0.422)	0.899 (0.871–0.925)	0.647 (0.638–0.656)
surgical_group	chopcluster_1	903	22	0.024	0.879	0.460	0.818	0.751
surgical_group	chopcluster_10	346	28	0.081	0.789 (0.673–0.889)	0.383 (0.233–0.594)	0.786 (0.625–0.933)	0.560 (0.505–0.614)
surgical_group	chopcluster_11	303	9	0.030	0.761	0.239	0.556	0.769
surgical_group	chopcluster_12	1623	62	0.038	0.856 (0.810–0.894)	0.192 (0.135–0.294)	0.919 (0.844–0.981)	0.654 (0.631–0.676)
surgical_group	chopcluster_13	213	43	0.202	0.796 (0.716–0.871)	0.561 (0.419–0.700)	0.930 (0.846–1.000)	0.253 (0.192–0.318)
surgical_group	chopcluster_3	807	22	0.027	0.941	0.482	0.955	0.753
surgical_group	chopcluster_4	3764	118	0.031	0.841 (0.793–0.883)	0.251 (0.188–0.338)	0.839 (0.767–0.902)	0.673 (0.658–0.687)

Attribute	Subgroup	n	Events	Prevalence	AUROC (95% CI)	AUPRC (95% CI)	Sensitivity (95% CI)	Specificity (95% CI)
surgical_group	chopcluster_5	769	30	0.039	0.893 (0.827–0.943)	0.392 (0.225–0.565)	0.933 (0.823–1.000)	0.636 (0.602–0.672)
surgical_group	chopcluster_6	530	67	0.126	0.889 (0.844–0.927)	0.625 (0.516–0.728)	0.955 (0.901–1.000)	0.471 (0.425–0.514)
surgical_group	chopcluster_7	298	36	0.121	0.864 (0.813–0.913)	0.521 (0.365–0.681)	1.000 (1.000–1.000)	0.309 (0.256–0.366)
surgical_group	chopcluster_8	306	23	0.075	0.802	0.327	0.870	0.551
surgical_group	none	857	44	0.051	0.900 (0.866–0.932)	0.388 (0.256–0.531)	0.977 (0.927–1.000)	0.625 (0.590–0.658)
emergency	Overall	10719	504	0.047	0.876 (0.859–0.892)	0.376 (0.332–0.422)	0.899 (0.871–0.925)	0.647 (0.638–0.656)
emergency	elective	9250	316	0.034	0.859 (0.837–0.879)	0.252 (0.209–0.303)	0.845 (0.808–0.883)	0.701 (0.692–0.711)
emergency	emergency	1469	188	0.128	0.858 (0.827–0.888)	0.554 (0.480–0.629)	0.989 (0.973–1.000)	0.268 (0.244–0.292)
sex × age_group	Overall	10719	504	0.047	0.876 (0.859–0.892)	0.376 (0.332–0.422)	0.899 (0.871–0.925)	0.647 (0.638–0.656)
sex × age_group	female / 0-17	175	13	0.063	0.902	0.408	0.907	0.549
sex × age_group	female / 18-44	408	13	0.038	0.873	0.340	0.896	0.663
sex × age_group	female / 45-64	1179	52	0.044	0.860 (0.788–0.919)	0.377 (0.248–0.517)	0.846 (0.738–0.941)	0.674 (0.647–0.702)
sex × age_group	female / 65-79	1470	93	0.063	0.900 (0.865–0.931)	0.545 (0.451–0.640)	0.935 (0.882–0.980)	0.645 (0.620–0.670)
sex × age_group	female / 80+	457	19	0.044	0.814	0.166	0.894	0.596
sex × age_group	male / 0-17	192	8	0.043	0.648	0.119	0.886	0.560
sex × age_group	male / 18-44	533	21	0.041	0.844	0.329	0.889	0.679
sex × age_group	male / 45-64	2536	95	0.037	0.903 (0.873–0.929)	0.392 (0.301–0.494)	0.958 (0.915–0.991)	0.662 (0.644–0.681)
sex × age_group	male / 65-79	3214	156	0.049	0.882 (0.848–0.911)	0.378 (0.307–0.463)	0.897 (0.843–0.942)	0.658 (0.641–0.675)
sex × age_group	male / 80+	555	34	0.057	0.798	0.269	0.891	0.595
sex × frailty	Overall	10719	504	0.047	0.876 (0.859–0.892)	0.376 (0.332–0.422)	0.899 (0.871–0.925)	0.647 (0.638–0.656)
sex × frailty	female / frail	1666	129	0.077	0.888 (0.855–0.918)	0.492 (0.408–0.577)	0.938 (0.892–0.977)	0.530 (0.505–0.556)
sex × frailty	female / non_frail	522	1	0.016	0.996	0.333	0.900	0.748
sex × frailty	female / pre_frail	1501	60	0.040	0.823 (0.761–0.877)	0.226 (0.148–0.345)	0.817 (0.710–0.907)	0.703 (0.679–0.726)
sex × frailty	male / frail	4473	271	0.061	0.874 (0.850–0.896)	0.395 (0.335–0.459)	0.923 (0.891–0.954)	0.604 (0.589–0.619)
sex × frailty	male / non_frail	500	5	0.020	0.857	0.071	0.896	0.785
sex × frailty	male / pre_frail	2057	38	0.021	0.816	0.169	0.873	0.696
age_group × frailty	Overall	10719	504	0.047	0.876 (0.859–0.892)	0.376 (0.332–0.422)	0.899 (0.871–0.925)	0.647 (0.638–0.656)
age_group × frailty	0-17 / frail	211	19	0.077	0.791	0.282	0.862	0.393
age_group × frailty	0-17 / non_frail	45	0	0.043	—	—	—	0.711
age_group × frailty	0-17 / pre_frail	111	2	0.039	0.539	0.042	0.852	0.510
age_group × frailty	18-44 / frail	300	21	0.060	0.864	0.463	0.835	0.565
age_group × frailty	18-44 / non_frail	331	2	0.015	0.869	0.031	0.826	0.789
age_group × frailty	18-44 / pre_frail	310	11	0.036	0.744	0.141	0.809	0.680

Attribute	Subgroup	n	Events	Prevalence	AUROC (95% CI)	AUPRC (95% CI)	Sensitivity (95% CI)	Specificity (95% CI)
age_group × frailty	45-64 / frail	1967	111	0.056	0.898 (0.869–0.926)	0.444 (0.354–0.540)	0.964 (0.927–0.992)	0.603 (0.581–0.625)
age_group × frailty	45-64 / non_frail	410	3	0.015	0.971	0.307	0.920	0.790
age_group × frailty	45-64 / pre_frail	1338	33	0.026	0.808	0.175	0.887	0.701
age_group × frailty	65-79 / frail	3032	204	0.067	0.889 (0.862–0.914)	0.487 (0.416–0.560)	0.922 (0.884–0.958)	0.610 (0.592–0.627)
age_group × frailty	65-79 / non_frail	218	1	0.023	0.592	0.010	0.905	0.742
age_group × frailty	65-79 / pre_frail	1434	44	0.033	0.870	0.247	0.905	0.714
age_group × frailty	80+ / frail	629	45	0.068	0.796	0.253	0.859	0.523
age_group × frailty	80+ / non_frail	18	0	0.046	—	—	—	0.667
age_group × frailty	80+ / pre_frail	365	8	0.030	0.740	0.209	0.836	0.652

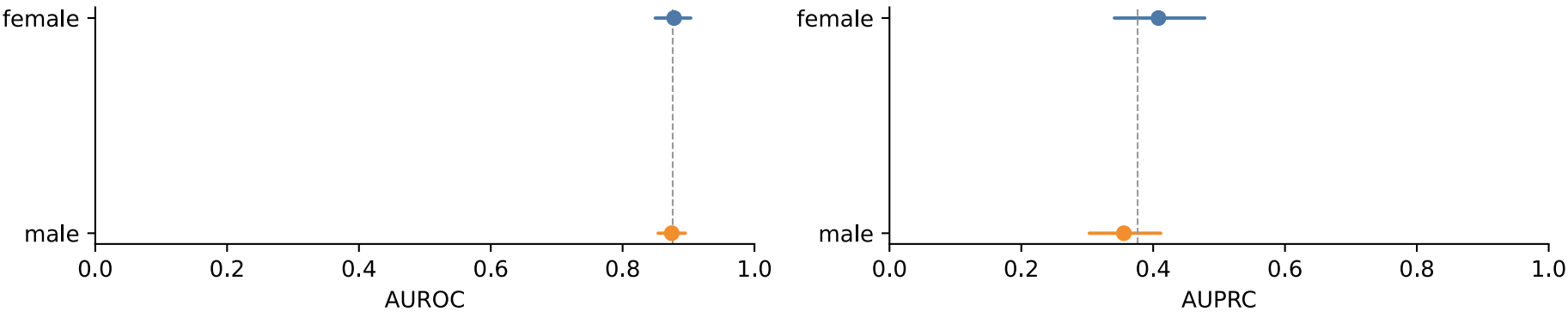
Gap statistics

Effect-size column shows Cohen's *h* on rate-metric gaps for two-group attributes; multi-group attributes and ranking metrics (AUROC / AUPRC) show "—".

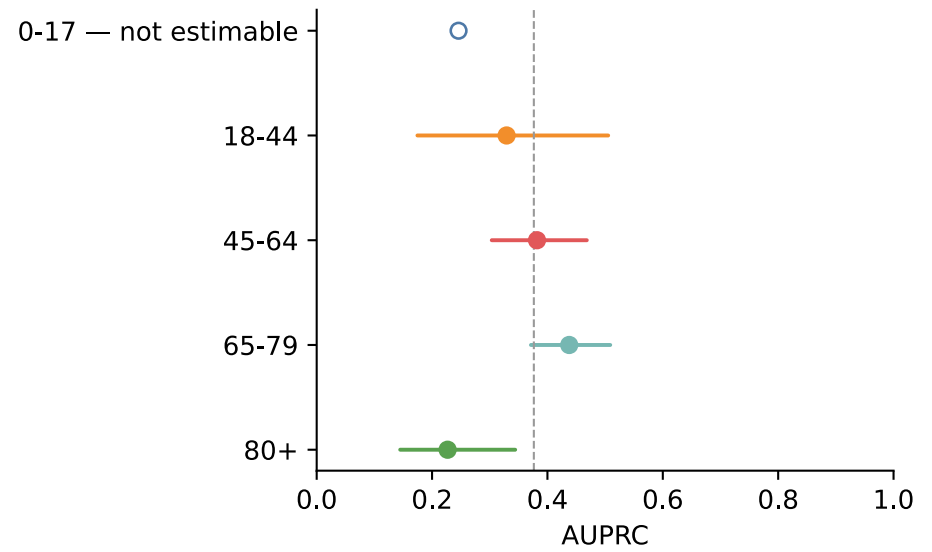
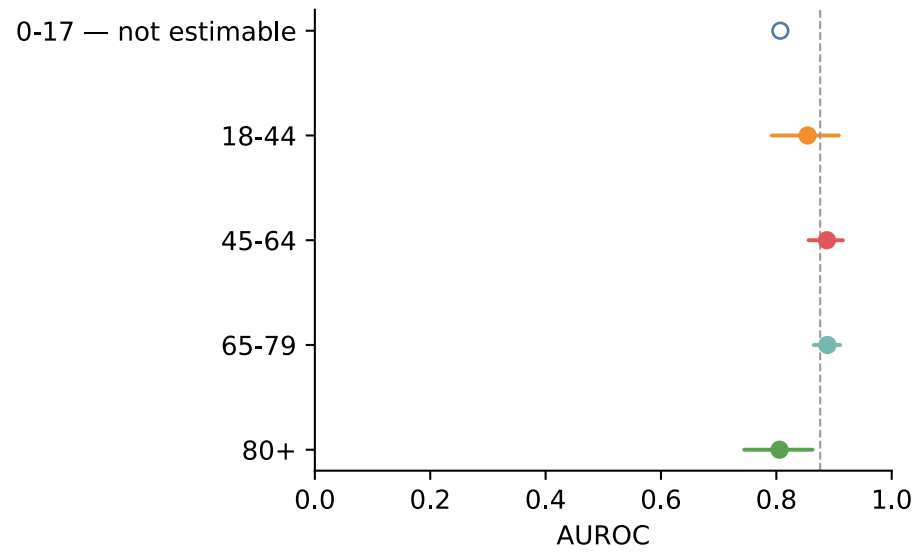
Attribute	Metric	Value	Effect size
sex	equal_opportunity_difference	0.0019	h=+0.01 (negligible)
sex	predictive_equality	0.0102	h=-0.02 (negligible)
sex	equalized_odds_gap	0.0102	—
sex	statistical_parity	0.0135	—
age_group	equal_opportunity_difference	0.0948	—
age_group	predictive_equality	0.2200	—
age_group	equalized_odds_gap	0.2200	—
age_group	statistical_parity	0.2200	—
frailty	equal_opportunity_difference	0.1418	—
frailty	predictive_equality	0.2320	—
frailty	equalized_odds_gap	0.2320	—
frailty	statistical_parity	0.2616	—
clinic	equal_opportunity_difference	0.3333	—
clinic	predictive_equality	0.3731	—
clinic	equalized_odds_gap	0.3731	—
clinic	statistical_parity	0.3857	—
surgical_group	equal_opportunity_difference	0.4444	—
surgical_group	predictive_equality	0.5158	—
surgical_group	equalized_odds_gap	0.5158	—

Attribute	Metric	Value	Effect size
surgical_group	statistical_parity	0.5431	—
emergency	equal_opportunity_difference	0.1444	h=−0.60 (medium)
emergency	predictive_equality	0.4335	h=+0.90 (large)
emergency	equalized_odds_gap	0.4335	—
emergency	statistical_parity	0.4477	—
sex × age_group	equal_opportunity_difference	0.1117	—
sex × age_group	predictive_equality	0.1300	—
sex × age_group	equalized_odds_gap	0.1300	—
sex × age_group	statistical_parity	0.1415	—
sex × frailty	equal_opportunity_difference	0.1213	—
sex × frailty	predictive_equality	0.2550	—
sex × frailty	equalized_odds_gap	0.2550	—
sex × frailty	statistical_parity	0.2844	—
age_group × frailty	equal_opportunity_difference	0.1546	—
age_group × frailty	predictive_equality	0.3970	—
age_group × frailty	equalized_odds_gap	0.3970	—
age_group × frailty	statistical_parity	0.4158	—

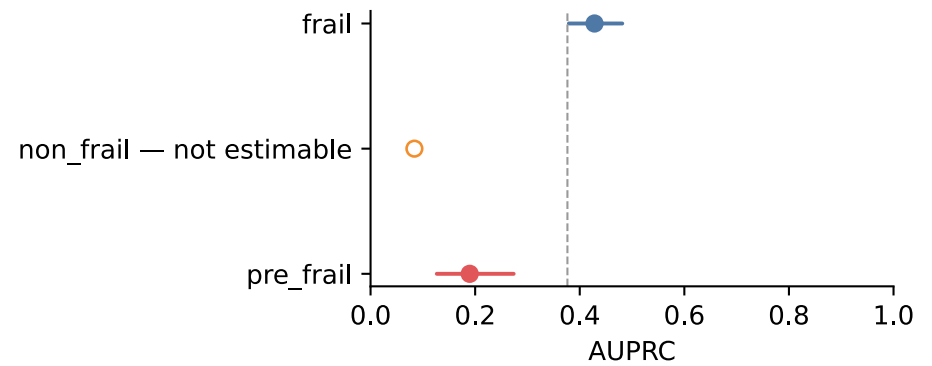
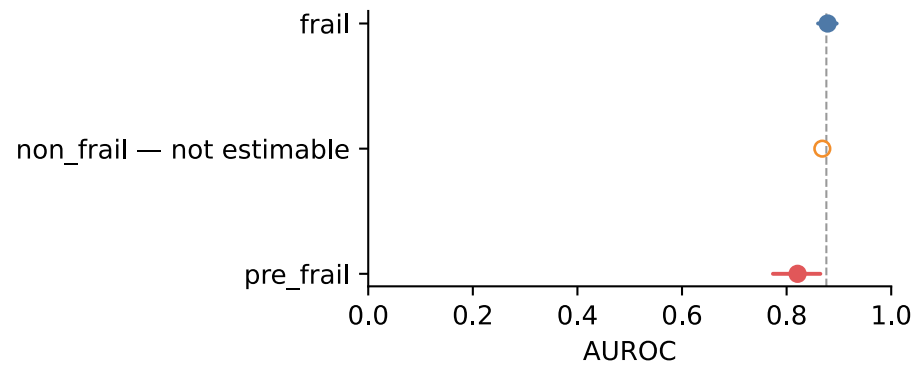
Discrimination — sex



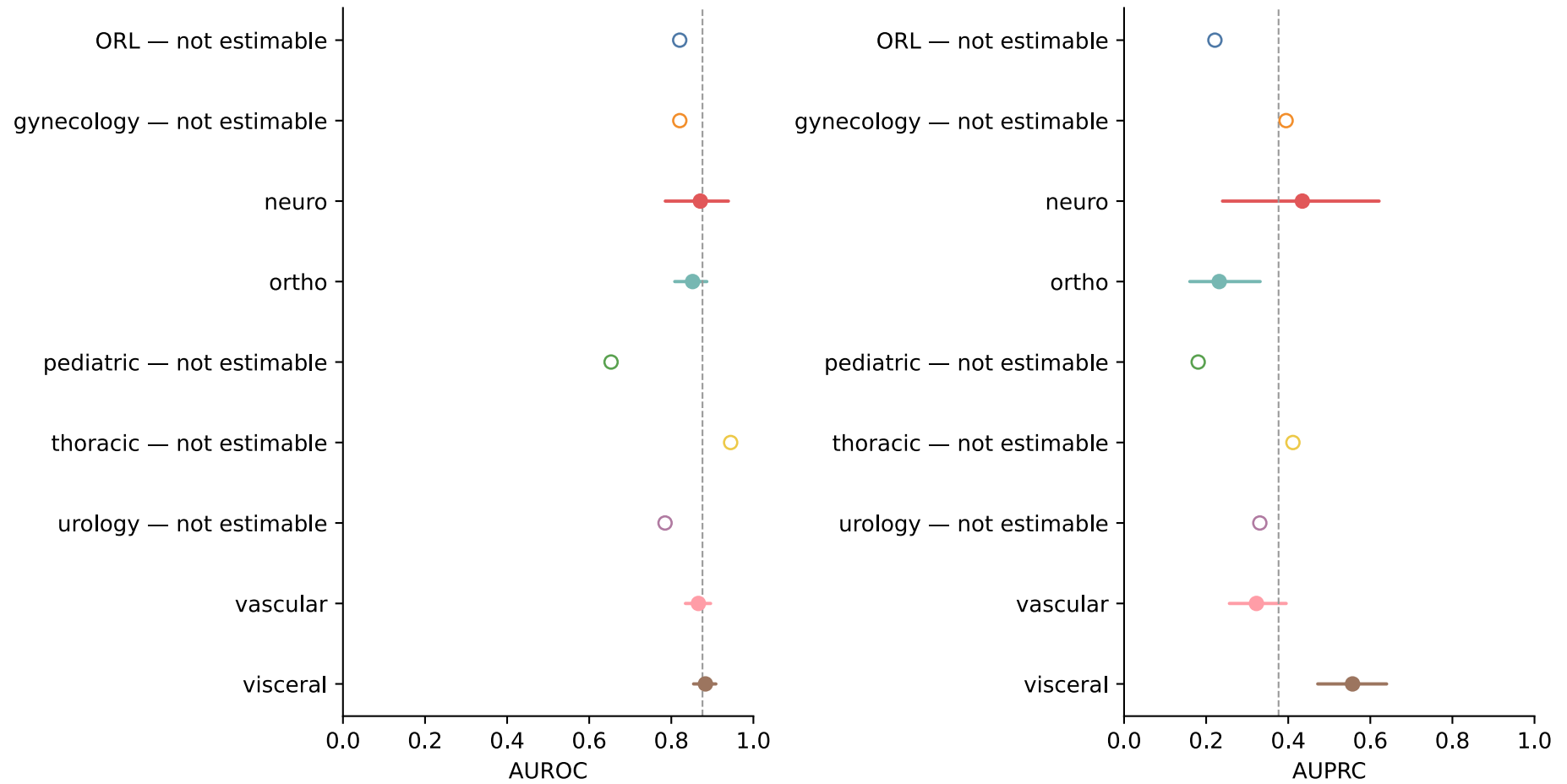
Discrimination — age_group



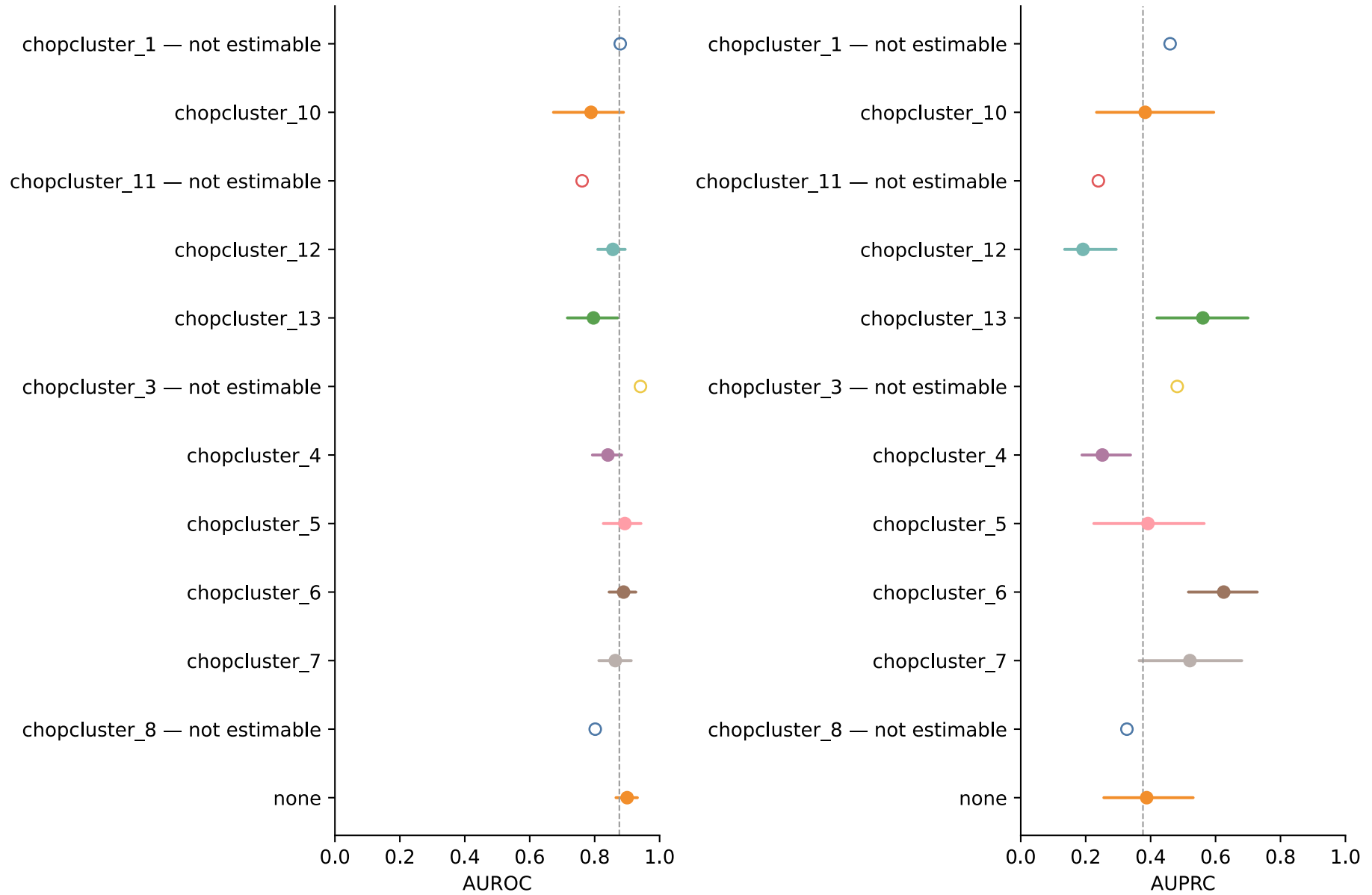
Discrimination — frailty



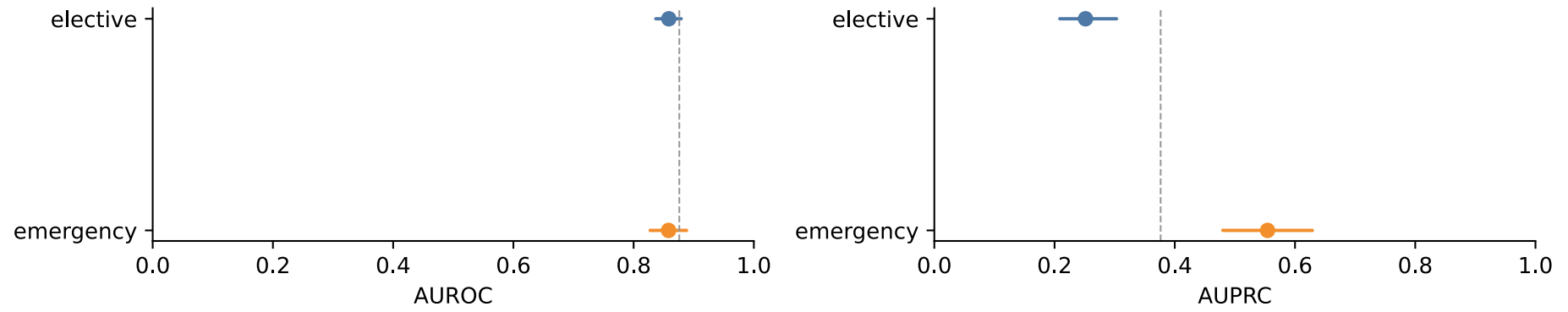
Discrimination — clinic



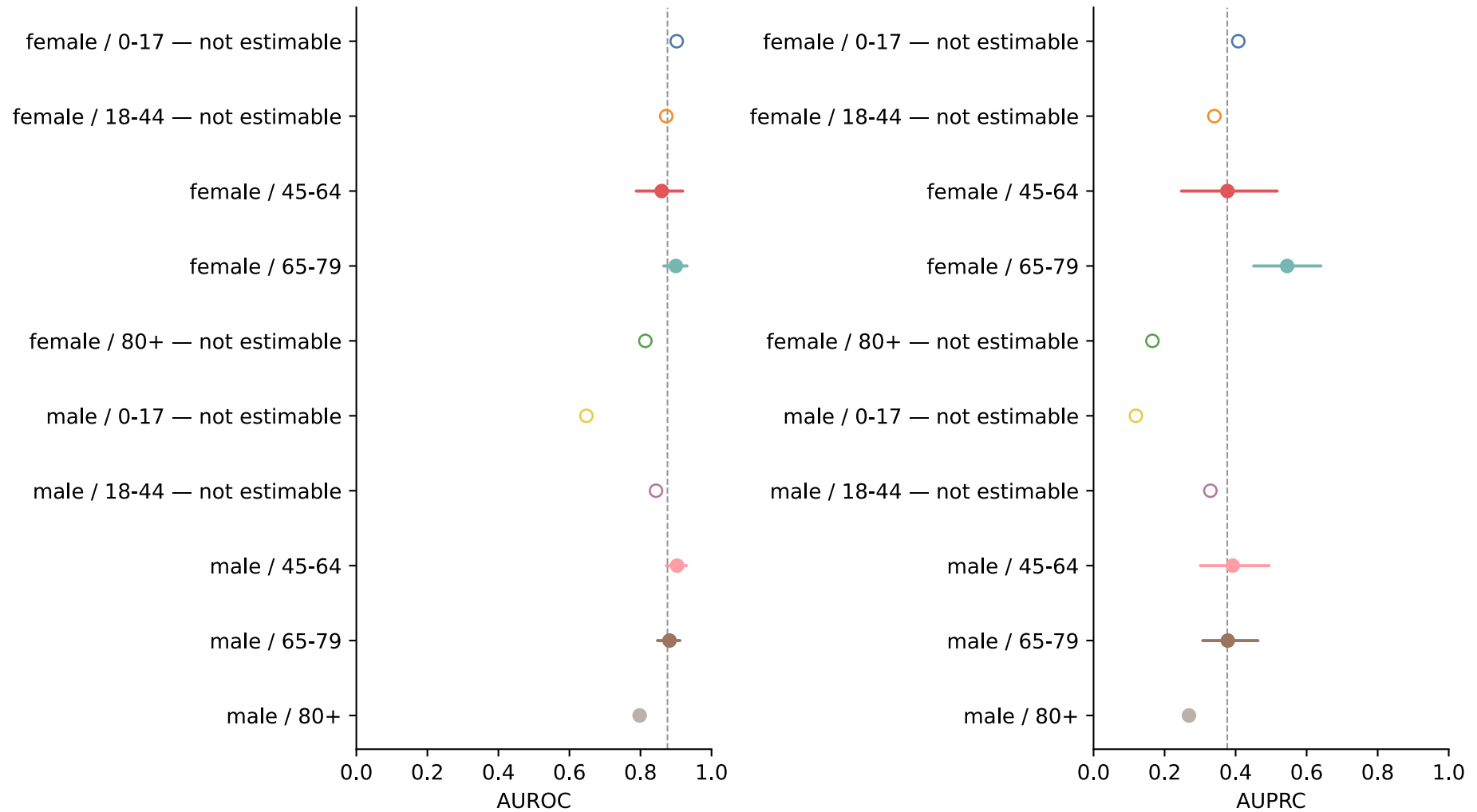
Discrimination — surgical_group



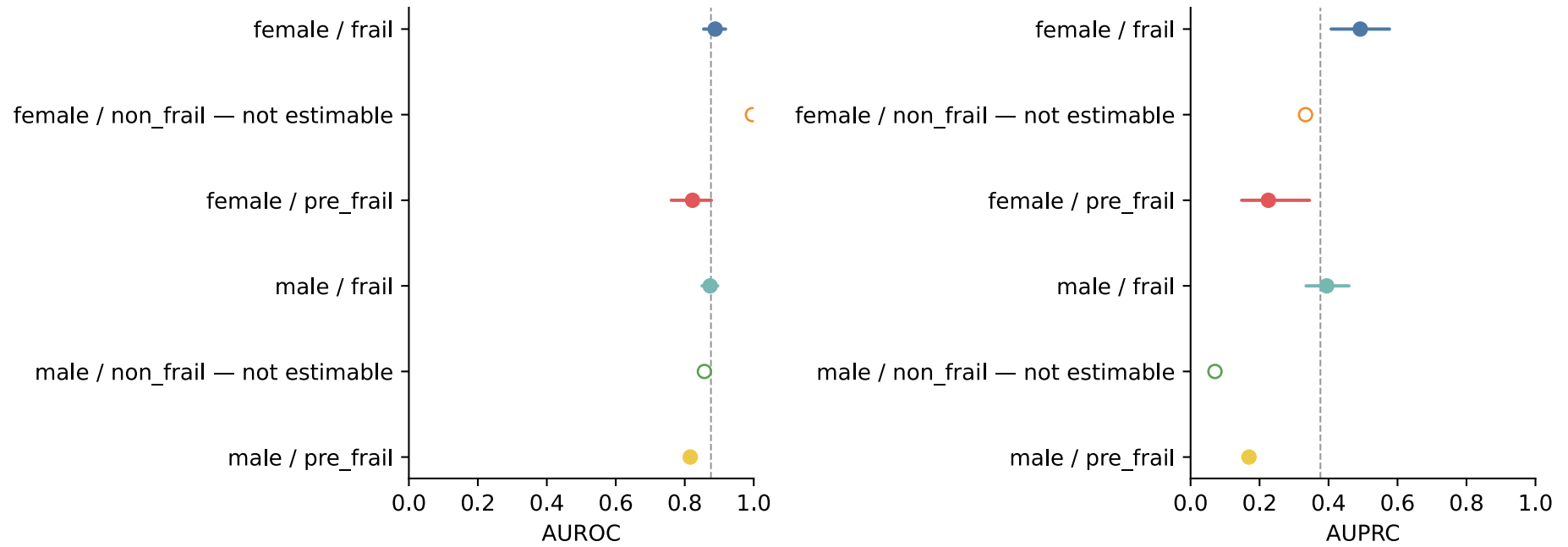
Discrimination — emergency

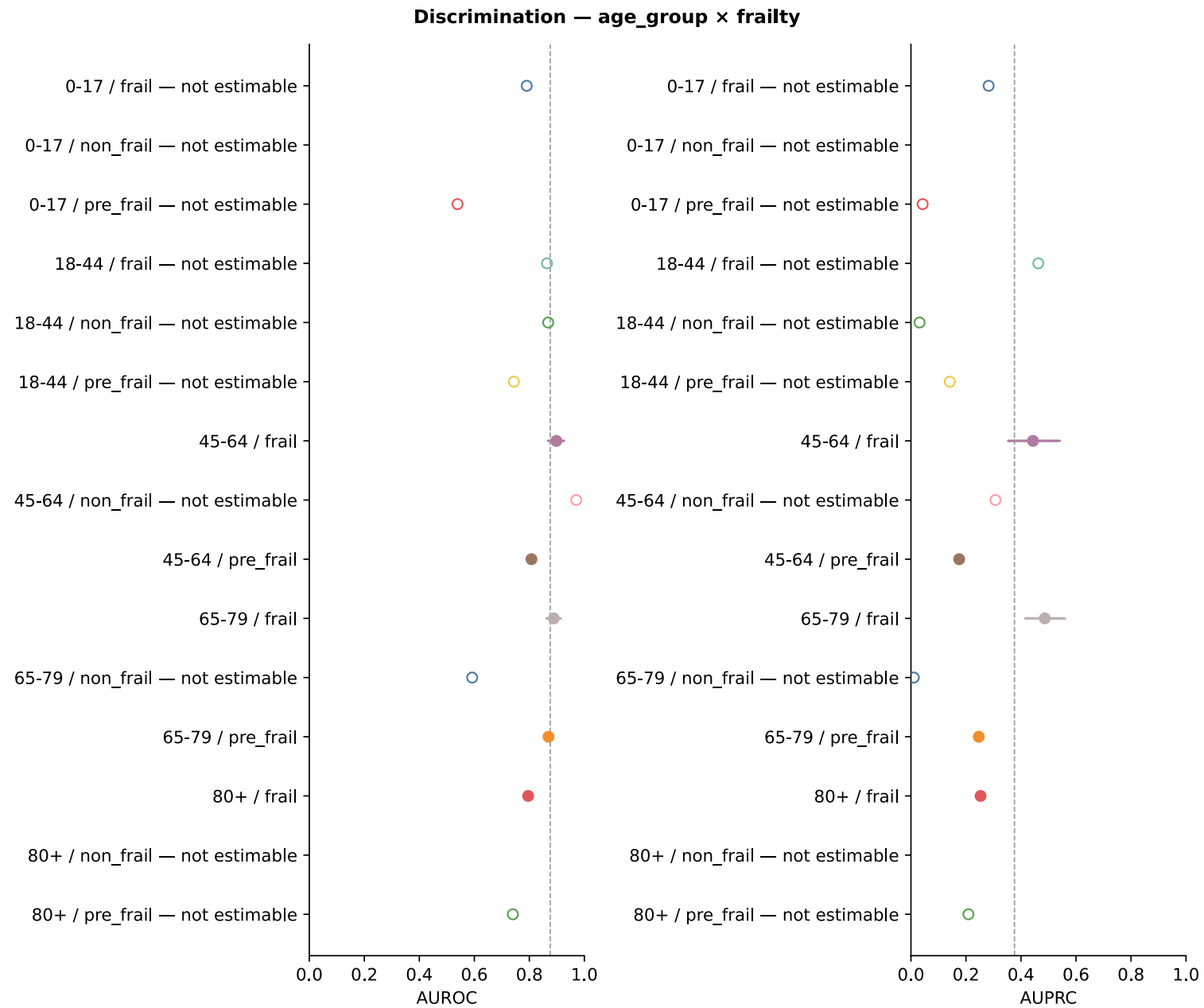


Discrimination — sex × age_group



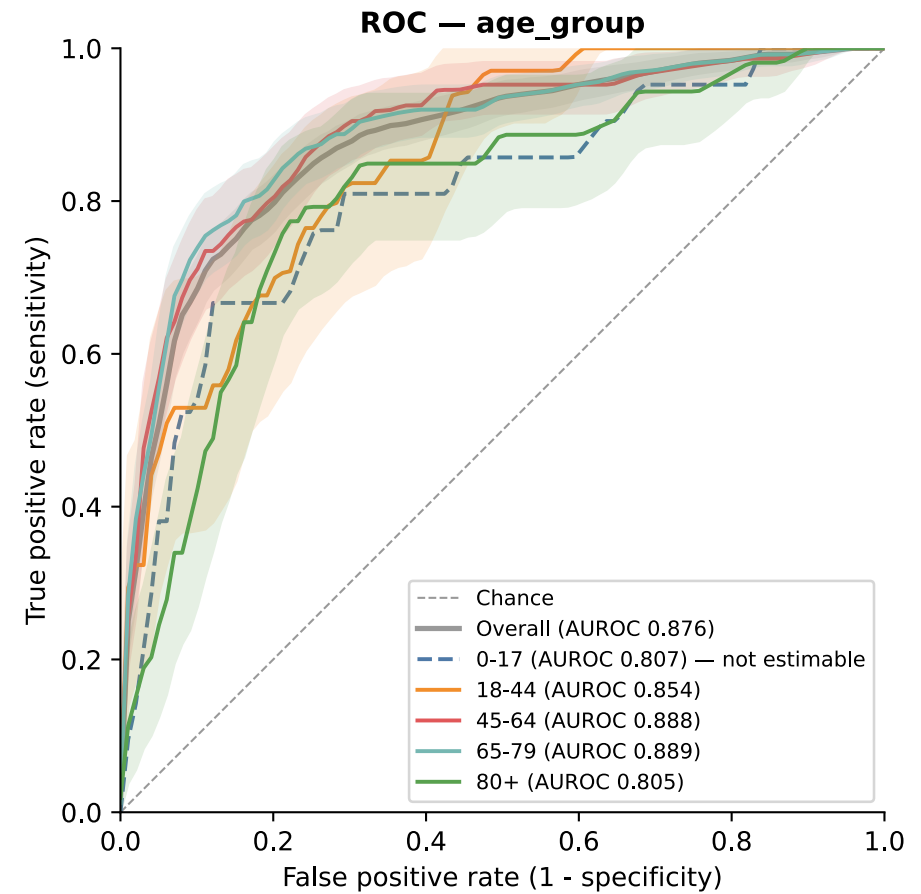
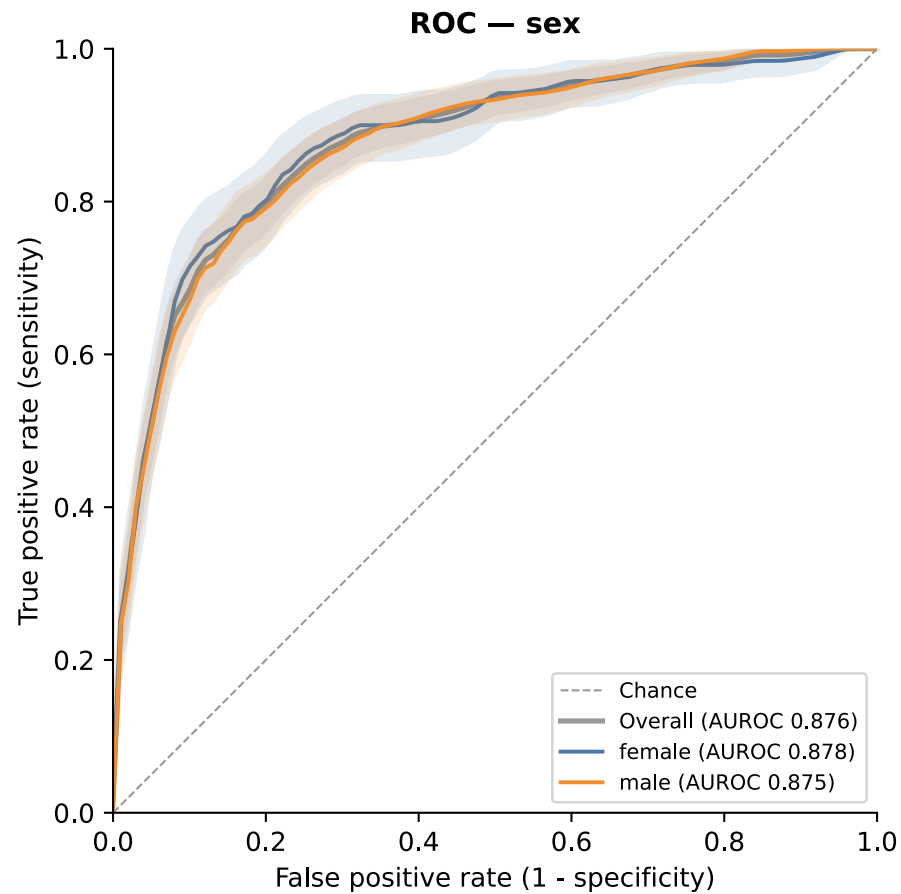
Discrimination — sex × frailty

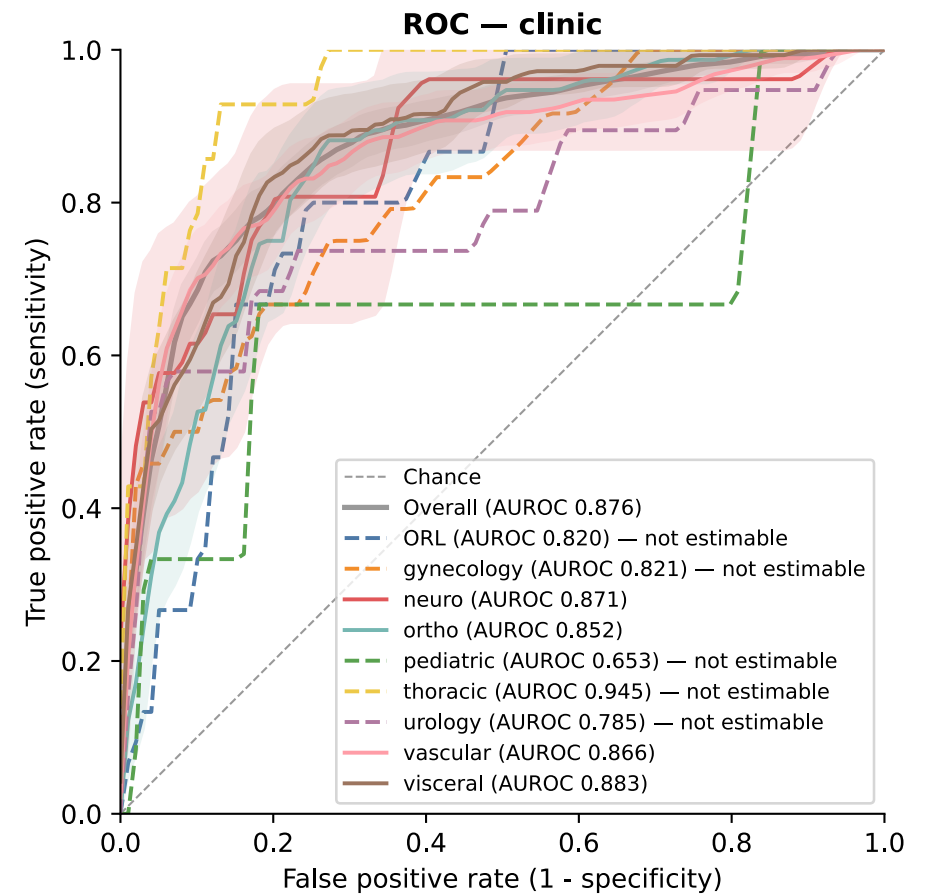
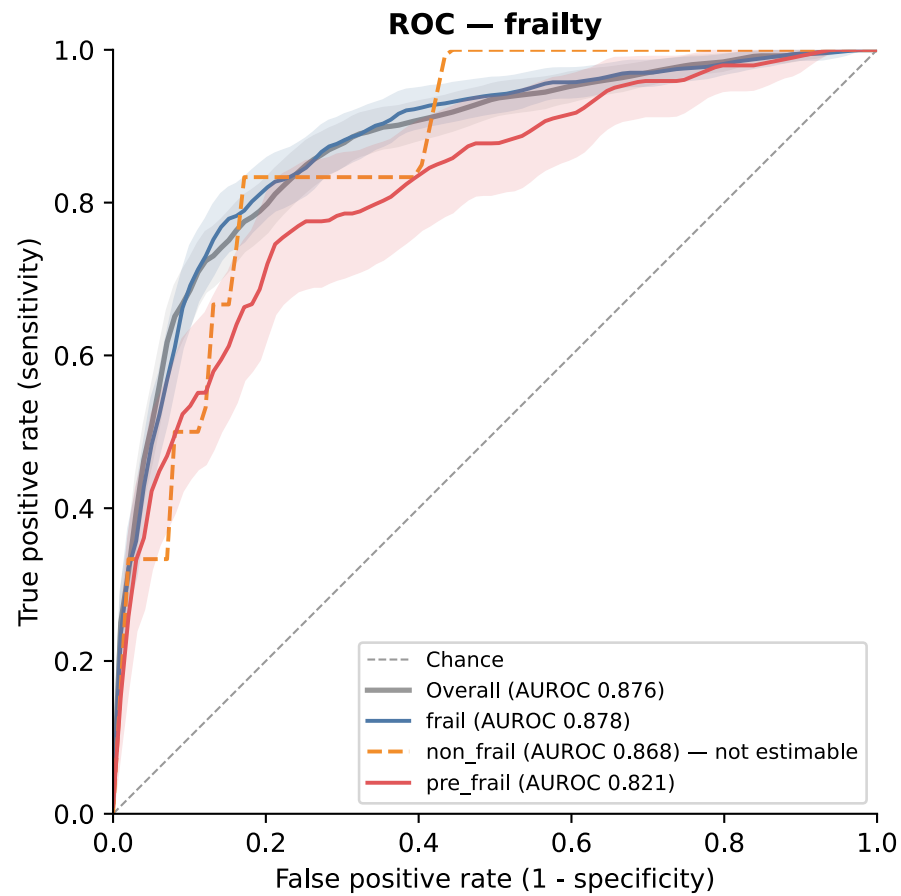


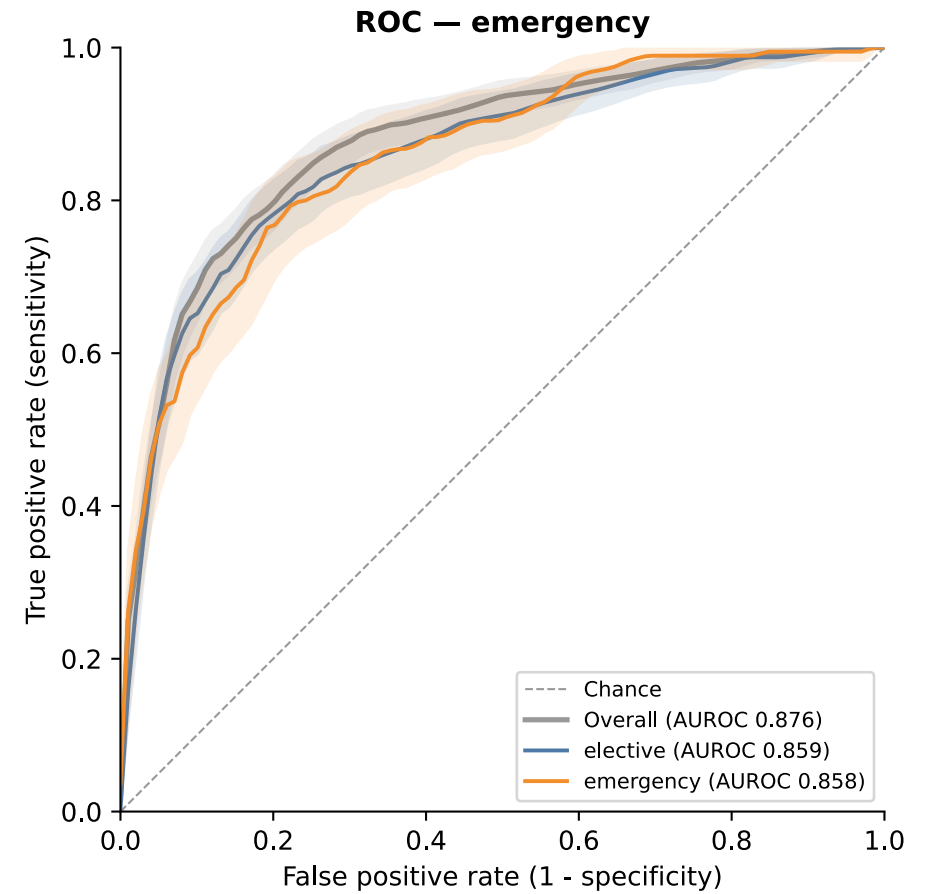
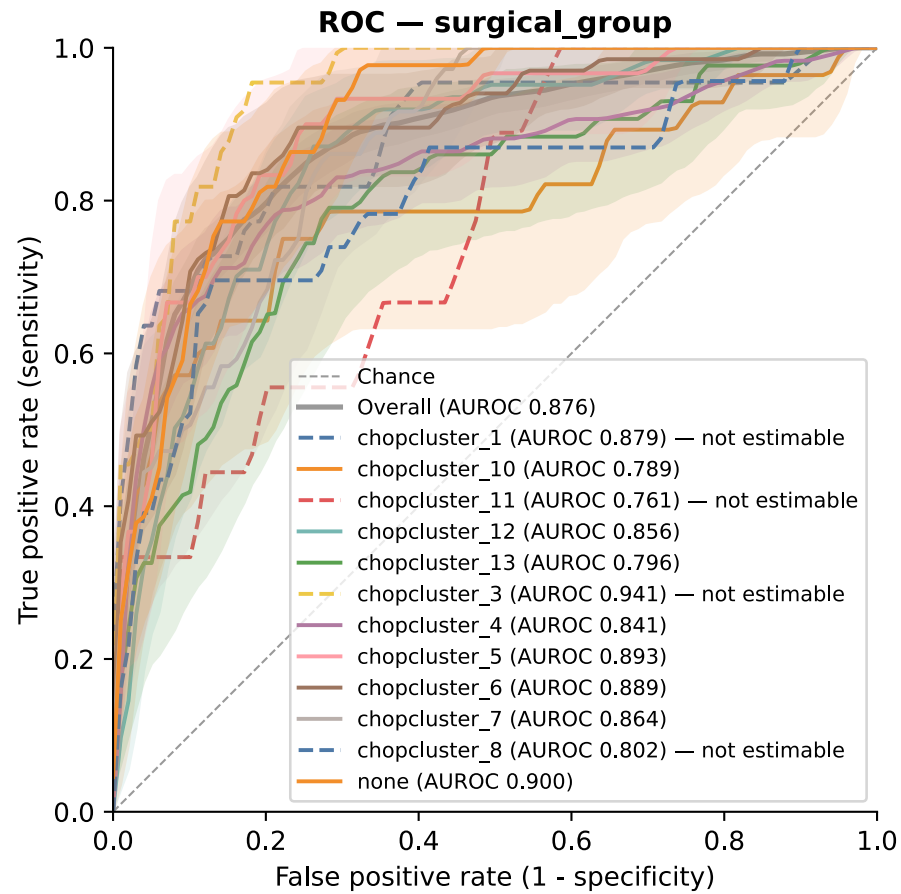


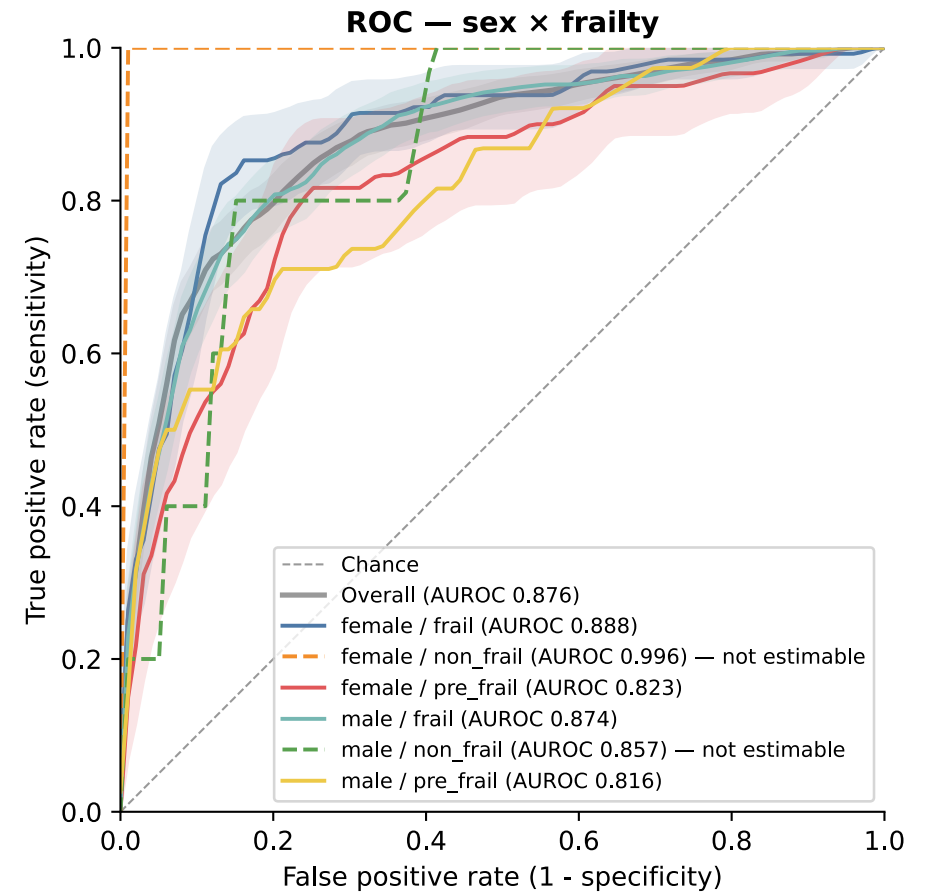
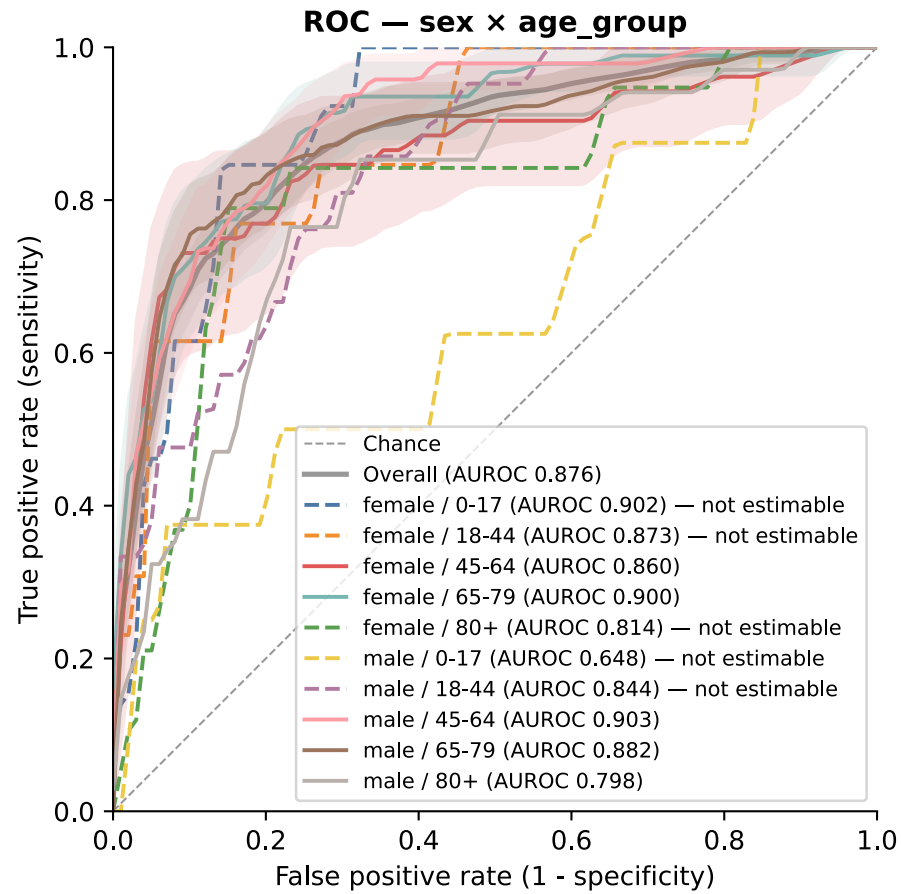
ROC curves

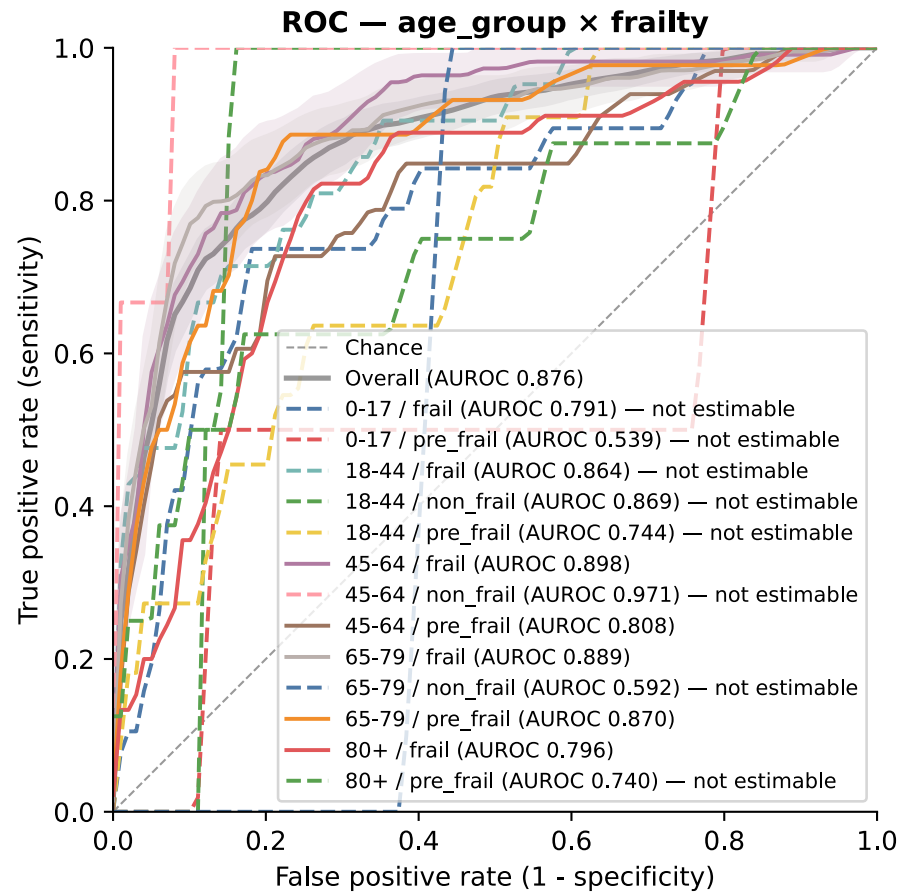
Per-subgroup ROC curves with pointwise 95% bootstrap bands. AUROC values in the legend are taken from the table above, not integrated from the drawn curve. Curves that cross indicate that no single subgroup dominates at every operating point — the ranking of subgroups depends on the threshold chosen.











Calibration

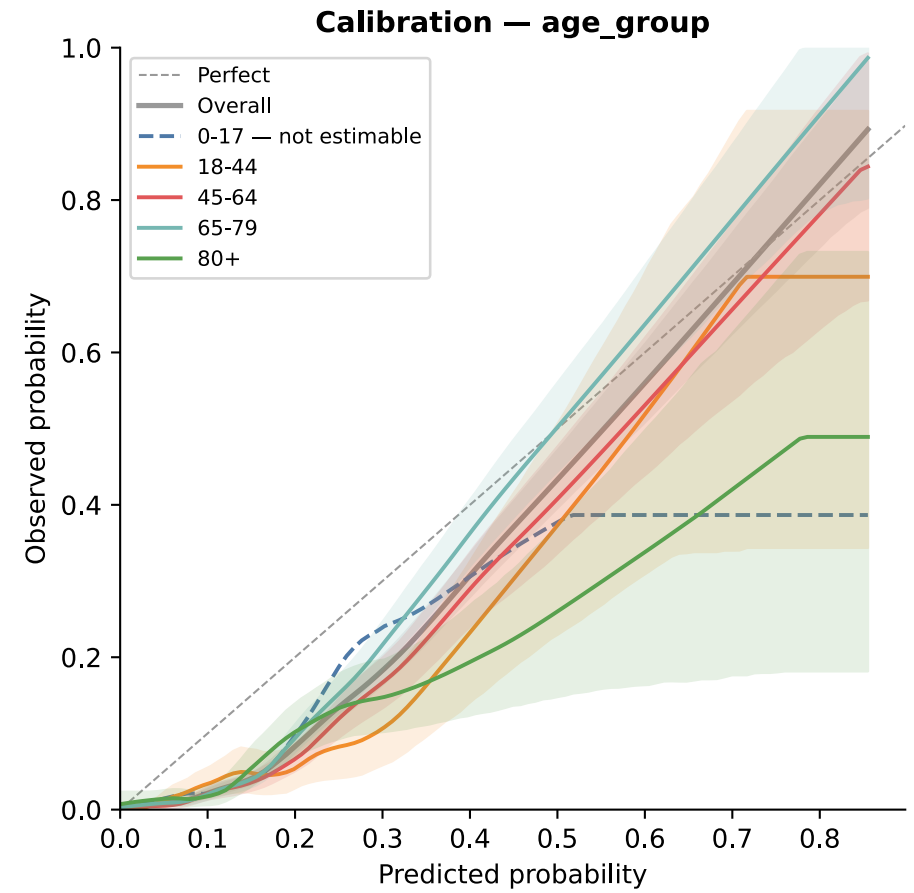
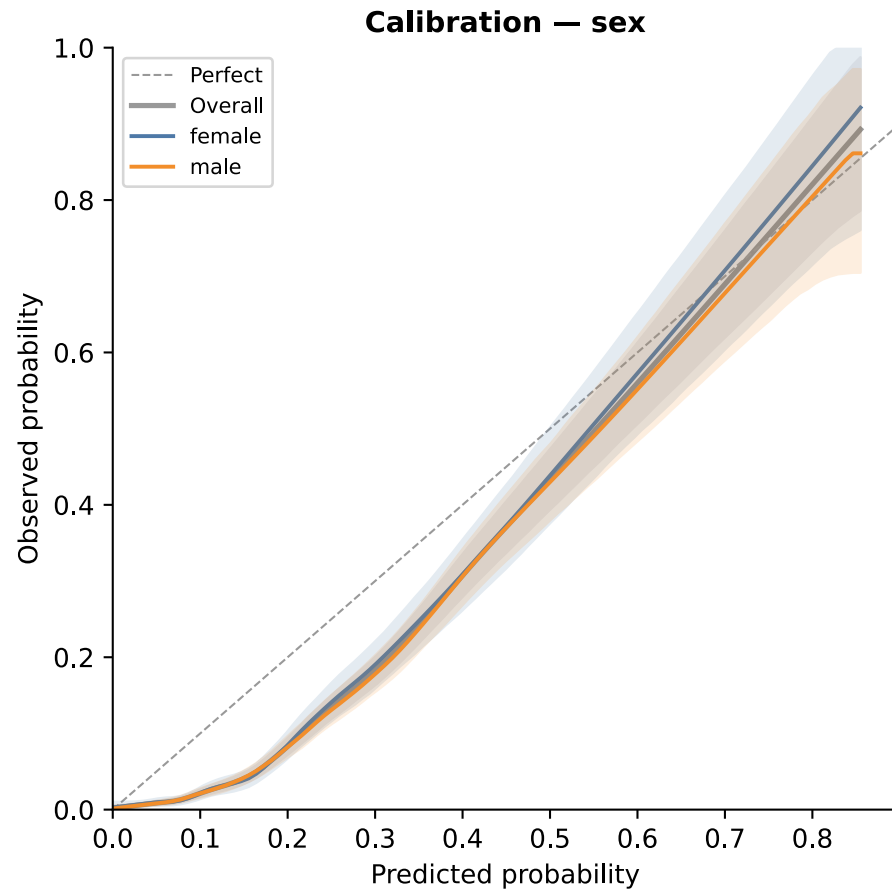
Attribute	Subgroup	n	Intercept (95% CI)	Slope (95% CI)	Brier (95% CI)	ECE	ICI
sex	Overall	10719	-1.020 (-1.116—-0.938)	1.566 (1.447–1.688)	0.041 (0.039–0.043)	0.0624	0.0609
sex	female	3689	-0.963 (-1.112—-0.821)	1.561 (1.370–1.762)	0.043 (0.040–0.047)	0.0614	0.0600
sex	male	7030	-1.053 (-1.171—-0.940)	1.568 (1.423–1.730)	0.040 (0.037–0.043)	0.0629	0.0614
age_group	Overall	10719	-1.020 (-1.116—-0.938)	1.566 (1.447–1.688)	0.041 (0.039–0.043)	0.0624	0.0609
age_group	0-17	367	-0.962	1.542	0.055	0.0733	0.0713
age_group	18-44	941	-1.207 (-1.590—-0.906)	1.404	0.036 (0.029–0.044)	0.0647	0.0636
age_group	45-64	3715	-1.162 (-1.343—-1.007)	1.594 (1.386–1.845)	0.036 (0.033–0.040)	0.0646	0.0631
age_group	65-79	4684	-0.889 (-1.020—-0.768)	1.678 (1.502–1.876)	0.042 (0.039–0.046)	0.0568	0.0560
age_group	80+	1012	-1.069 (-1.376—-0.810)	1.164	0.053 (0.046–0.062)	0.0745	0.0748
frailty	Overall	10719	-1.020 (-1.116—-0.938)	1.566 (1.447–1.688)	0.041 (0.039–0.043)	0.0624	0.0609
frailty	frail	6139	-0.855 (-0.959—-0.757)	1.594 (1.463–1.741)	0.052 (0.049–0.055)	0.0637	0.0620
frailty	non_frail	1022	-2.475	1.546	0.011	0.0550	0.0550
frailty	pre_frail	3558	-1.342 (-1.554—-1.158)	1.282	0.031 (0.028–0.034)	0.0623	0.0617
clinic	Overall	10719	-1.020 (-1.116—-0.938)	1.566 (1.447–1.688)	0.041 (0.039–0.043)	0.0624	0.0609
clinic	ORL	233	-1.150	1.302	0.068	0.0984	0.0982
clinic	gynecology	452	-0.692	1.414	0.044	0.0423	0.0450
clinic	neuro	1117	-1.306 (-1.769—-0.963)	1.858	0.022 (0.017–0.028)	0.0522	0.0536
clinic	ortho	1690	-1.073 (-1.331—-0.866)	1.325	0.045 (0.039–0.051)	0.0654	0.0649
clinic	pediatric	65	-1.218	0.710	0.054	0.1069	0.0887
clinic	thoracic	723	-1.542	1.982	0.020	0.0560	0.0552
clinic	urology	349	-0.776	1.286	0.048	0.0546	0.0510
clinic	vascular	4852	-1.147 (-1.297—-1.008)	1.657 (1.443–1.895)	0.036 (0.033–0.039)	0.0639	0.0624
clinic	visceral	1238	-0.655 (-0.825—-0.479)	1.572 (1.339–1.849)	0.078 (0.071–0.086)	0.0713	0.0669
surgical_group	Overall	10719	-1.020 (-1.116—-0.938)	1.566 (1.447–1.688)	0.041 (0.039–0.043)	0.0624	0.0609
surgical_group	chopcluster_1	903	-1.323	1.834	0.023	0.0546	0.0551
surgical_group	chopcluster_10	346	-0.493 (-0.954—-0.138)	1.360	0.064 (0.047–0.082)	0.0599	0.0516
surgical_group	chopcluster_11	303	-1.111	1.123	0.030	0.0533	0.0502
surgical_group	chopcluster_12	1623	-1.174 (-1.447—-0.941)	1.339	0.040 (0.034–0.046)	0.0658	0.0650
surgical_group	chopcluster_13	213	-0.112 (-0.451—0.206)	1.377	0.127 (0.103–0.156)	0.0579	0.0323
surgical_group	chopcluster_3	807	-1.351	1.937	0.026	0.0587	0.0581
surgical_group	chopcluster_4	3764	-1.266 (-1.468—-1.093)	1.583 (1.306–1.889)	0.032 (0.029–0.035)	0.0644	0.0630

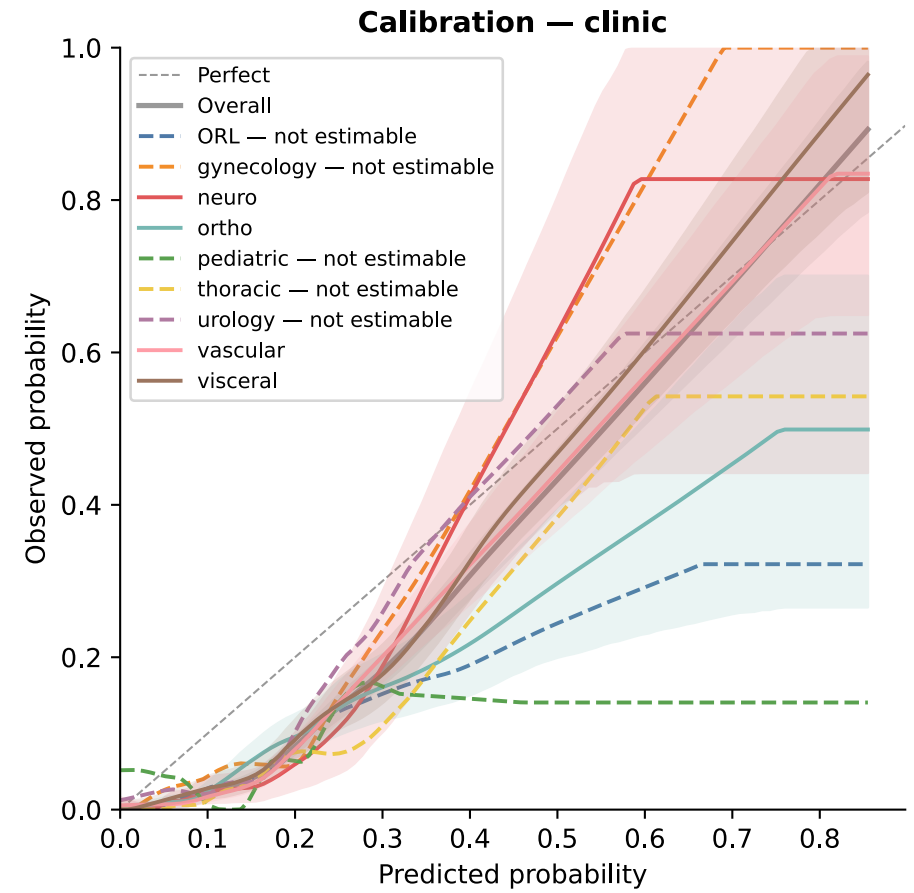
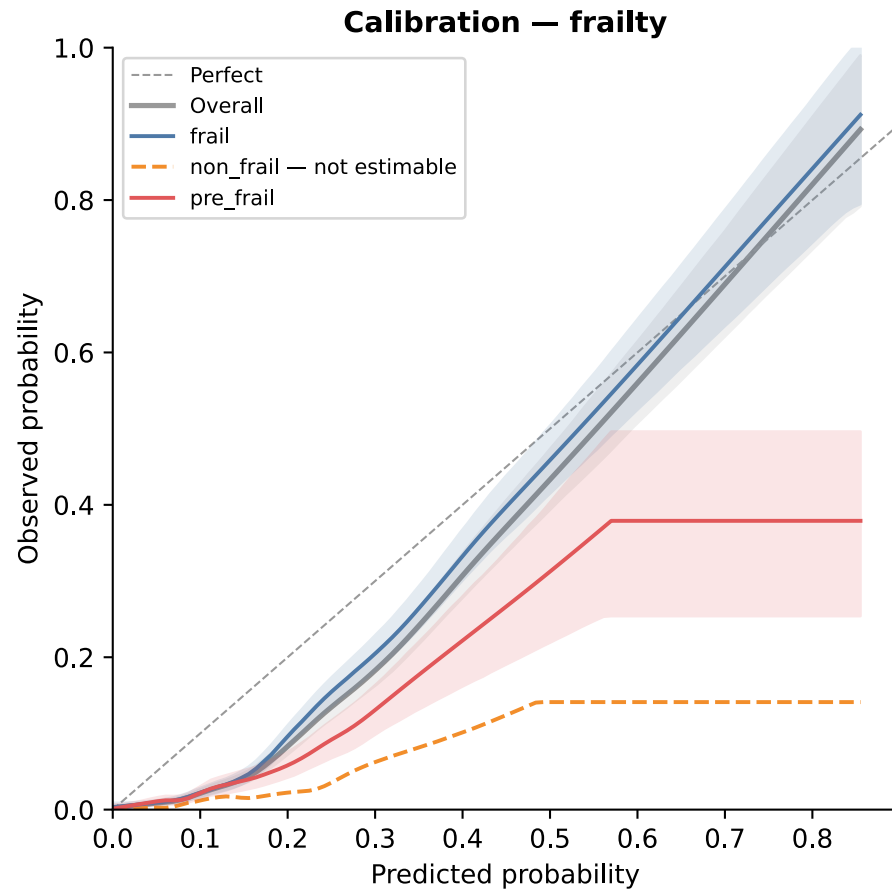
Attribute	Subgroup	n	Intercept (95% CI)	Slope (95% CI)	Brier (95% CI)	ECE	ICI
surgical_group	chopcluster_5	769	-1.201 (-1.605—0.877)	1.727	0.036 (0.029–0.044)	0.0684	0.0670
surgical_group	chopcluster_6	530	-0.565 (-0.838—0.316)	1.621	0.078 (0.066–0.091)	0.0665	0.0646
surgical_group	chopcluster_7	298	-0.788 (-1.178—0.462)	1.752	0.090 (0.074–0.105)	0.0932	0.0928
surgical_group	chopcluster_8	306	-0.903	1.085	0.068	0.0823	0.0739
surgical_group	none	857	-1.006 (-1.367—0.717)	1.582	0.045 (0.038–0.053)	0.0644	0.0632
emergency	Overall	10719	-1.020 (-1.116—0.938)	1.566 (1.447–1.688)	0.041 (0.039–0.043)	0.0624	0.0609
emergency	elective	9250	-1.156 (-1.270—1.046)	1.488 (1.351–1.630)	0.034 (0.031–0.036)	0.0588	0.0577
emergency	emergency	1469	-0.705 (-0.860—0.561)	1.727 (1.493–2.006)	0.089 (0.082–0.097)	0.0936	0.0878
sex × age_group	Overall	10719	-1.020 (-1.116—0.938)	1.566 (1.447–1.688)	0.041 (0.039–0.043)	0.0624	0.0609
sex × age_group	female / 0-17	175	-0.761	2.236	0.061	0.0771	0.0644
sex × age_group	female / 18-44	408	-1.395	1.530	0.034	0.0731	0.0720
sex × age_group	female / 45-64	1179	-1.058 (-1.378—0.785)	1.355	0.039 (0.032–0.046)	0.0606	0.0591
sex × age_group	female / 65-79	1470	-0.750 (-0.973—0.546)	1.805	0.046 (0.040–0.053)	0.0519	0.0556
sex × age_group	female / 80+	457	-1.310	1.179	0.048	0.0834	0.0839
sex × age_group	male / 0-17	192	-1.217	0.785	0.049	0.0859	0.0802
sex × age_group	male / 18-44	533	-1.072	1.334	0.038	0.0582	0.0578
sex × age_group	male / 45-64	2536	-1.213 (-1.419—1.023)	1.752	0.035 (0.031–0.039)	0.0666	0.0654
sex × age_group	male / 65-79	3214	-0.962 (-1.125—0.809)	1.612 (1.403–1.863)	0.041 (0.037–0.045)	0.0590	0.0572
sex × age_group	male / 80+	555	-0.904	1.168	0.058	0.0671	0.0676
sex × frailty	Overall	10719	-1.020 (-1.116—0.938)	1.566 (1.447–1.688)	0.041 (0.039–0.043)	0.0624	0.0609
sex × frailty	female / frail	1666	-0.819 (-1.006—0.654)	1.683 (1.436–1.985)	0.058 (0.052–0.065)	0.0683	0.0672
sex × frailty	female / non_frail	522	-3.691	4.400	0.009	0.0637	0.0638
sex × frailty	female / pre_frail	1501	-0.983 (-1.257—0.737)	1.226	0.039 (0.033–0.046)	0.0531	0.0525
sex × frailty	male / frail	4473	-0.872 (-0.999—0.762)	1.561 (1.397–1.737)	0.050 (0.046–0.053)	0.0620	0.0601
sex × frailty	male / non_frail	500	-1.834	1.374	0.014	0.0460	0.0459
sex × frailty	male / pre_frail	2057	-1.730	1.331	0.025	0.0691	0.0686
age_group × frailty	Overall	10719	-1.020 (-1.116—0.938)	1.566 (1.447–1.688)	0.041 (0.039–0.043)	0.0624	0.0609
age_group × frailty	0-17 / frail	211	-0.689	1.442	0.079	0.0705	0.0664
age_group × frailty	0-17 / non_frail	45	—	—	0.007	0.0721	0.0721
age_group × frailty	0-17 / pre_frail	111	-1.857	0.242	0.028	0.0927	0.0844
age_group × frailty	18-44 / frail	300	-0.990	1.603	0.060	0.0827	0.0811
age_group × frailty	18-44 / non_frail	331	-2.473	1.062	0.013	0.0561	0.0562
age_group × frailty	18-44 / pre_frail	310	-1.078	0.965	0.039	0.0564	0.0567

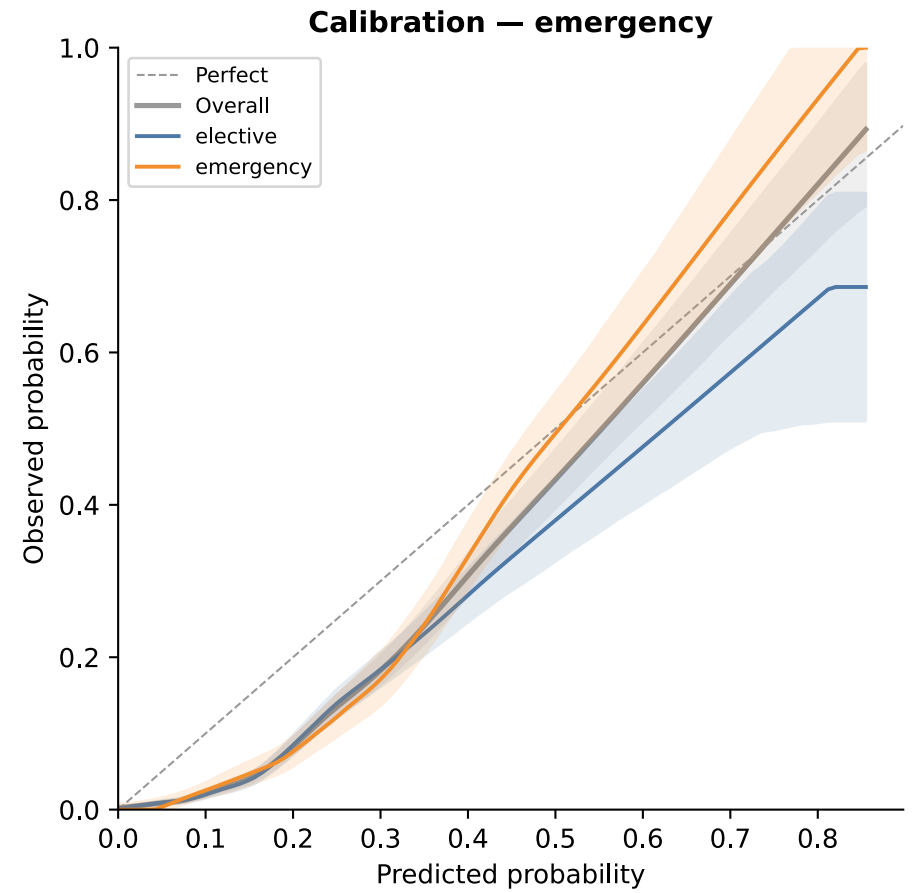
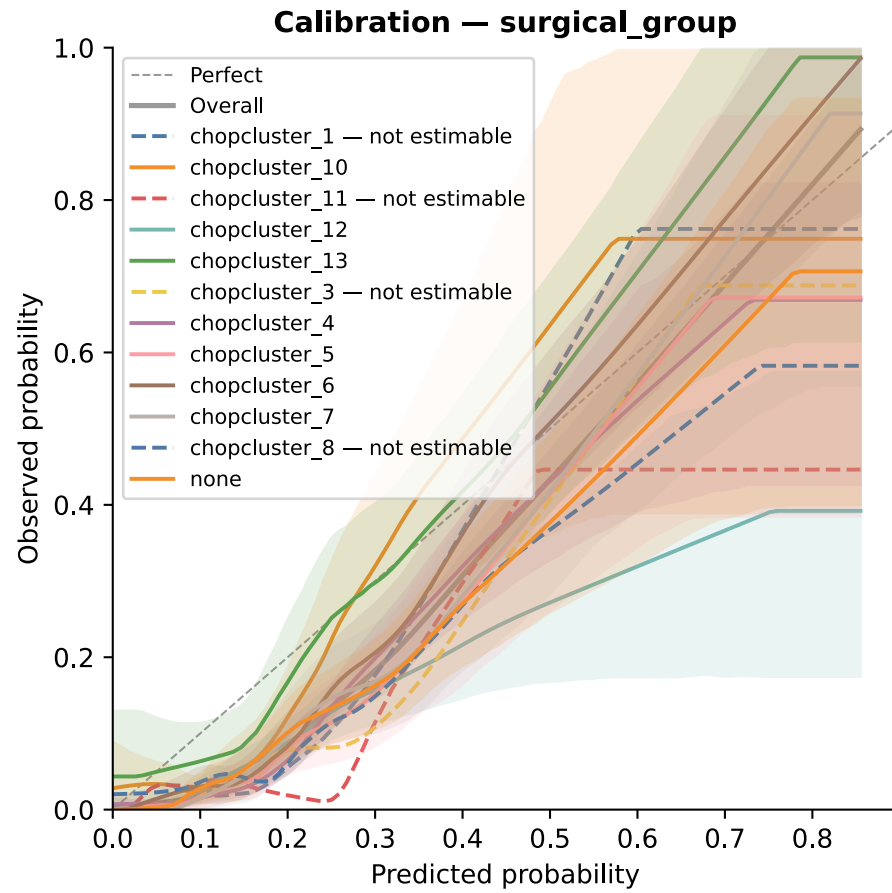
Attribute	Subgroup	n	Intercept (95% CI)	Slope (95% CI)	Brier (95% CI)	ECE	ICI
age_group × frailty	45-64 / frail	1967	-1.000 (-1.198—-0.827)	1.633 (1.403–1.921)	0.046 (0.041–0.051)	0.0687	0.0667
age_group × frailty	45-64 / non_frail	410	-2.215	2.787	0.011	0.0512	0.0510
age_group × frailty	45-64 / pre_frail	1338	-1.420	1.252	0.029	0.0625	0.0619
age_group × frailty	65-79 / frail	3032	-0.758 (-0.907—-0.629)	1.727 (1.522–1.976)	0.050 (0.045–0.054)	0.0568	0.0575
age_group × frailty	65-79 / non_frail	218	-2.693	0.196	0.011	0.0552	0.0553
age_group × frailty	65-79 / pre_frail	1434	-1.240	1.444	0.032	0.0593	0.0583
age_group × frailty	80+ / frail	629	-0.899	1.126	0.068	0.0753	0.0763
age_group × frailty	80+ / non_frail	18	—	—	0.012	0.0783	0.0783
age_group × frailty	80+ / pre_frail	365	-1.635	0.975	0.030	0.0739	0.0726

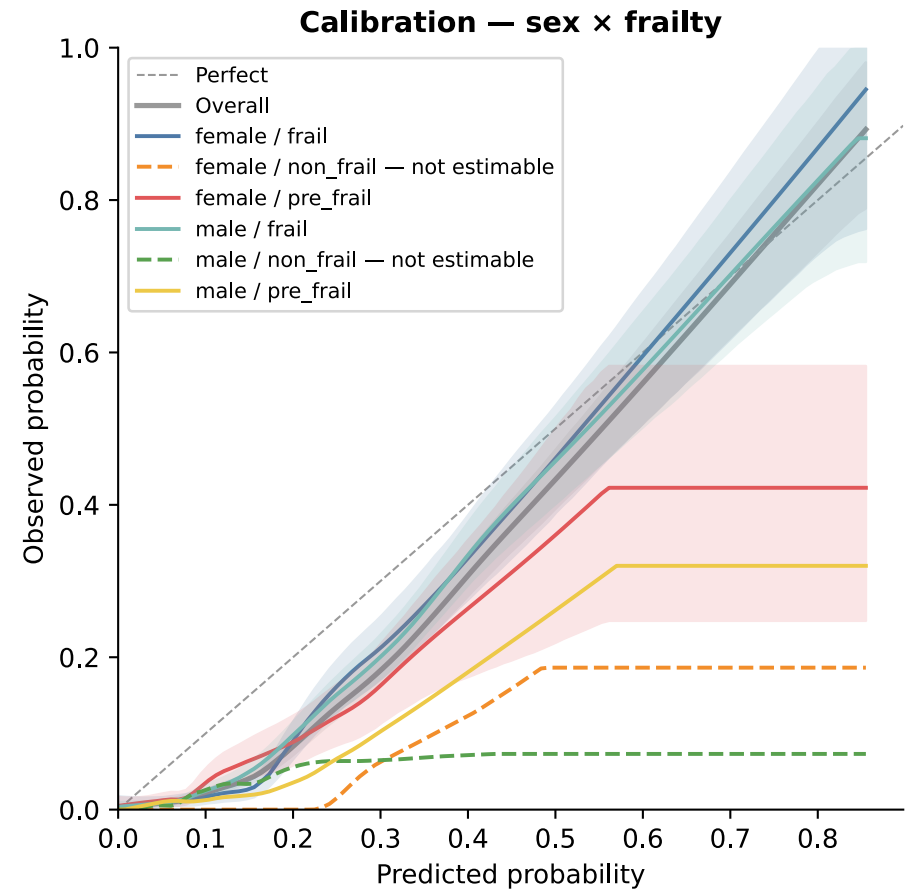
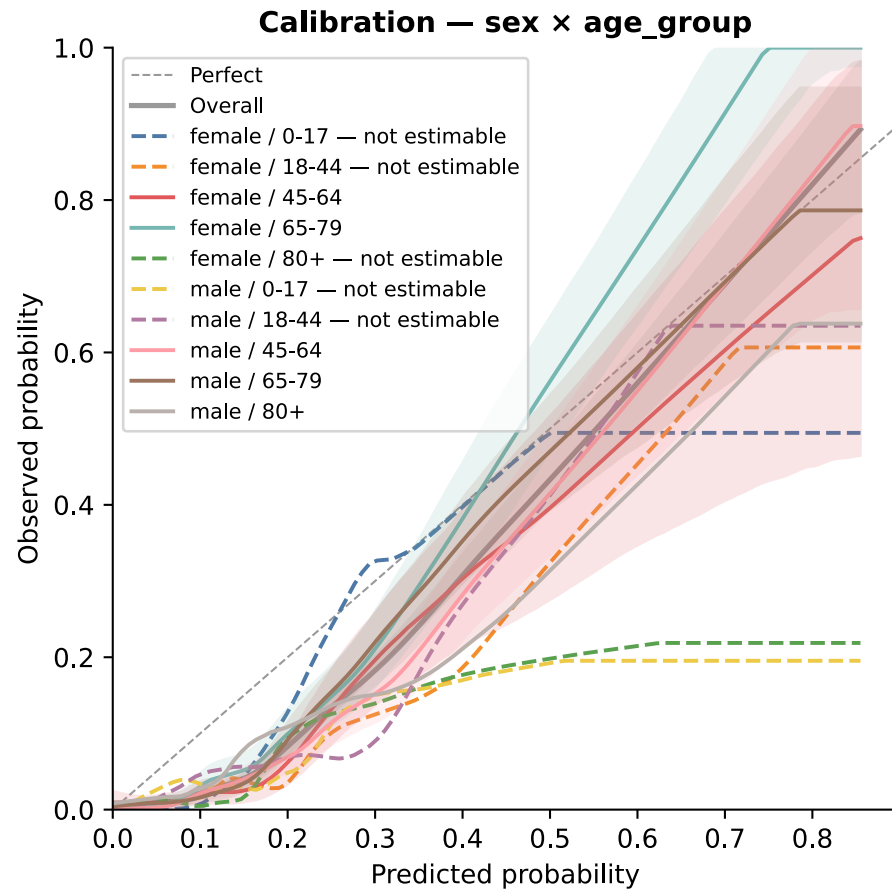
Calibration gap statistics

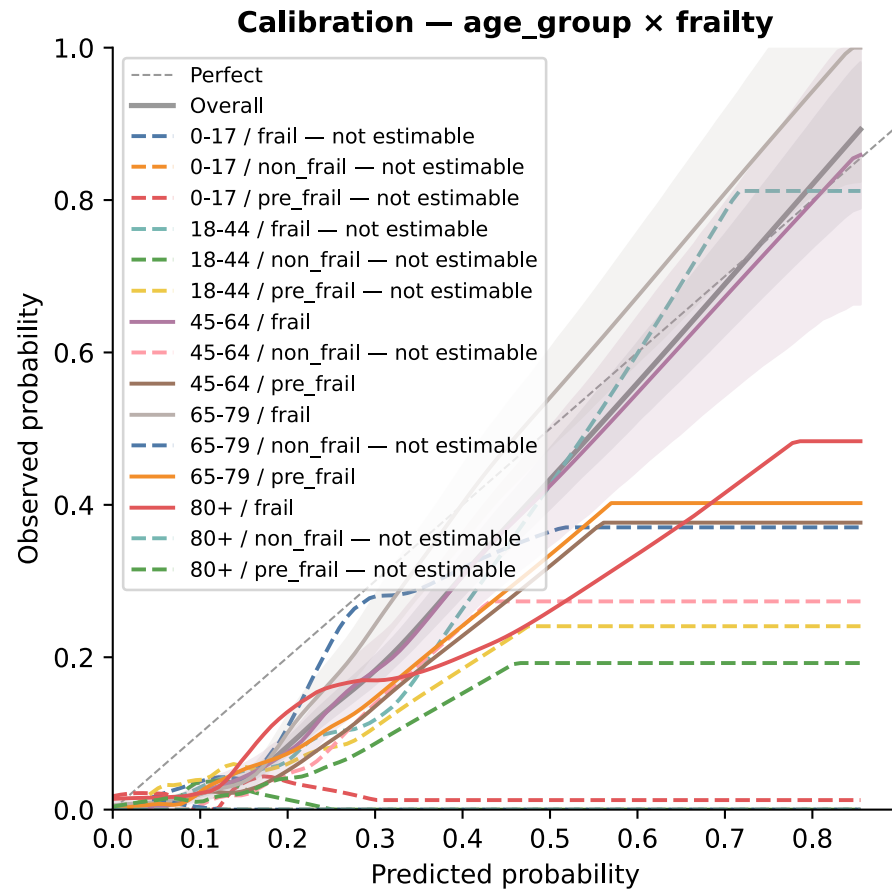
Attribute	Metric	Value (95% CI)
sex	max_intercept_gap	0.0905 (0.0048–0.2827)
sex	max_slope_gap	0.0066 (0.0038–0.2911)
age_group	max_intercept_gap	0.3186
age_group	max_slope_gap	0.5136
frailty	max_intercept_gap	1.6196
frailty	max_slope_gap	0.3126
clinic	max_intercept_gap	0.8870
clinic	max_slope_gap	1.2723
surgical_group	max_intercept_gap	1.2385
surgical_group	max_slope_gap	0.8526
emergency	max_intercept_gap	0.4515 (0.2610–0.6342)
emergency	max_slope_gap	0.2381 (0.0209–0.5473)
sex × age_group	max_intercept_gap	0.6449
sex × age_group	max_slope_gap	1.4507
sex × frailty	max_intercept_gap	2.8720
sex × frailty	max_slope_gap	3.1740
age_group × frailty	max_intercept_gap	2.0037
age_group × frailty	max_slope_gap	2.5912











Hosmer–Lemeshow goodness-of-fit

Read every statistic beside its *n* and its decile table. The Hosmer–Lemeshow statistic scales with subgroup size: a clinically irrelevant deviation becomes significant in a large subgroup, while a large deviation can go undetected in a small one. Values are therefore **not comparable across subgroups of different size**, and subgroups must not be ranked by p-value. This is why Kramer & Zimmerman (2007) require the statistic to be reported together with the number of patients, the per-decile observed-versus-expected table below, and the adjunct calibration measures in the table above — never on its own.

Attribute	Subgroup	n	Events	H–L statistic	df	p	Groups used	Decile slope	Decile intercept
sex	Overall	10719	504	531.10	10	0.0000	10	0.746	-0.035
sex	female	3689	190	177.76	10	0.0000	10	0.777	-0.037
sex	male	7030	314	352.91	10	0.0000	10	0.718	-0.033
age_group	Overall	10719	504	531.10	10	0.0000	10	0.746	-0.035

Attribute	Subgroup	n	Events	H–L statistic	df	p	Groups used	Decile slope	Decile intercept
age_group	0-17	367	21	22.54	10	0.0126	10	0.833	-0.050
age_group	18-44	941	34	50.47	10	0.0000	10	0.532	-0.018
age_group	45-64	3715	147	201.77	10	0.0000	10	0.675	-0.031
age_group	65-79	4684	249	220.84	10	0.0000	10	0.884	-0.044
age_group	80+	1012	53	60.51	10	0.0000	10	0.535	-0.016
frailty	Overall	10719	504	531.10	10	0.0000	10	0.746	-0.035
frailty	frail	6139	400	293.55	10	0.0000	10	0.860	-0.046
frailty	non_frail	1022	6	58.09	10	0.0000	10	0.150	-0.003
frailty	pre_frail	3558	98	187.73	10	0.0000	10	0.436	-0.011
clinic	Overall	10719	504	531.10	10	0.0000	10	0.746	-0.035
clinic	ORL	233	15	22.35	10	0.0134	10	0.537	-0.023
clinic	gynecology	452	24	13.57	10	0.1937	10	0.783	-0.021
clinic	neuro	1117	26	50.71	10	0.0000	10	0.561	-0.020
clinic	ortho	1690	76	86.86	10	0.0000	10	0.593	-0.021
clinic	pediatric	65	3	7.93	10	0.6359	10	0.335	-0.001
clinic	thoracic	723	14	37.02	10	0.0001	10	0.477	-0.016
clinic	urology	349	19	17.83	10	0.0578	10	0.856	-0.036
clinic	vascular	4852	184	263.12	10	0.0000	10	0.701	-0.034
clinic	visceral	1238	143	66.58	10	0.0000	10	0.988	-0.066
surgical_group	Overall	10719	504	531.10	10	0.0000	10	0.746	-0.035
surgical_group	chopcluster_1	903	22	44.33	10	0.0000	10	0.575	-0.022
surgical_group	chopcluster_10	346	28	15.64	10	0.1105	10	1.125	-0.056
surgical_group	chopcluster_11	303	9	13.28	10	0.2085	10	0.339	+0.003
surgical_group	chopcluster_12	1623	62	86.61	10	0.0000	10	0.563	-0.020
surgical_group	chopcluster_13	213	43	5.89	10	0.8241	10	1.242	-0.070
surgical_group	chopcluster_3	807	22	42.67	10	0.0000	10	0.604	-0.025
surgical_group	chopcluster_4	3764	118	209.96	10	0.0000	10	0.608	-0.027
surgical_group	chopcluster_5	769	30	46.17	10	0.0000	10	0.690	-0.034
surgical_group	chopcluster_6	530	67	25.39	10	0.0046	10	1.076	-0.074
surgical_group	chopcluster_7	298	36	23.18	10	0.0101	10	0.990	-0.092
surgical_group	chopcluster_8	306	23	21.05	10	0.0207	10	0.653	-0.023
surgical_group	none	857	44	43.12	10	0.0000	10	0.695	-0.030
emergency	Overall	10719	504	531.10	10	0.0000	10	0.746	-0.035

Attribute	Subgroup	n	Events	H–L statistic	df	p	Groups used	Decile slope	Decile intercept
emergency	elective	9250	316	441.97	10	0.0000	10	0.635	-0.025
emergency	emergency	1469	188	104.02	10	0.0000	10	1.123	-0.112
sex × age_group	Overall	10719	504	531.10	10	0.0000	10	0.746	-0.035
sex × age_group	female / 0-17	175	13	13.85	10	0.1799	10	1.176	-0.091
sex × age_group	female / 18-44	408	13	27.47	10	0.0022	10	0.542	-0.025
sex × age_group	female / 45-64	1179	52	60.90	10	0.0000	10	0.708	-0.030
sex × age_group	female / 65-79	1470	93	61.58	10	0.0000	10	0.946	-0.046
sex × age_group	female / 80+	457	19	37.70	10	0.0000	10	0.466	-0.017
sex × age_group	male / 0-17	192	8	15.38	10	0.1188	10	0.388	-0.008
sex × age_group	male / 18-44	533	21	24.01	10	0.0076	10	0.521	-0.011
sex × age_group	male / 45-64	2536	95	143.70	10	0.0000	10	0.648	-0.030
sex × age_group	male / 65-79	3214	156	159.99	10	0.0000	10	0.834	-0.041
sex × age_group	male / 80+	555	34	27.49	10	0.0022	10	0.590	-0.014
sex × frailty	Overall	10719	504	531.10	10	0.0000	10	0.746	-0.035
sex × frailty	female / frail	1666	129	95.15	10	0.0000	10	0.915	-0.056
sex × frailty	female / non_frail	522	1	37.09	10	0.0001	10	0.077	-0.003
sex × frailty	female / pre_frail	1501	60	57.99	10	0.0000	10	0.580	-0.014
sex × frailty	male / frail	4473	271	207.44	10	0.0000	10	0.837	-0.042
sex × frailty	male / non_frail	500	5	21.78	10	0.0162	10	0.247	-0.004
sex × frailty	male / pre_frail	2057	38	135.81	10	0.0000	10	0.312	-0.009
age_group × frailty	Overall	10719	504	531.10	10	0.0000	10	0.746	-0.035
age_group × frailty	0-17 / frail	211	19	12.10	10	0.2786	10	0.942	-0.058
age_group × frailty	0-17 / non_frail	45	0	—	—	—	10	—	—
age_group × frailty	0-17 / pre_frail	111	2	11.79	10	0.2992	10	-0.020	+0.021
age_group × frailty	18-44 / frail	300	21	19.58	10	0.0335	10	0.721	-0.041
age_group × frailty	18-44 / non_frail	331	2	20.32	10	0.0264	10	0.078	+0.001
age_group × frailty	18-44 / pre_frail	310	11	14.85	10	0.1377	10	0.353	+0.004
age_group × frailty	45-64 / frail	1967	111	109.75	10	0.0000	10	0.803	-0.044
age_group × frailty	45-64 / non_frail	410	3	21.21	10	0.0197	10	0.315	-0.011
age_group × frailty	45-64 / pre_frail	1338	33	77.02	10	0.0000	10	0.419	-0.011
age_group × frailty	65-79 / frail	3032	204	139.86	10	0.0000	10	0.985	-0.054
age_group × frailty	65-79 / non_frail	218	1	13.51	10	0.1968	10	-0.015	+0.005
age_group × frailty	65-79 / pre_frail	1434	44	71.06	10	0.0000	10	0.529	-0.017

Attribute	Subgroup	n	Events	H–L statistic	df	p	Groups used	Decile slope	Decile intercept
<i>age_group × frailty</i>	<i>80+ / frail</i>	629	45	40.56	10	0.0000	10	0.592	-0.016
<i>age_group × frailty</i>	<i>80+ / non_frail</i>	18	0	—	—	—	10	—	—
<i>age_group × frailty</i>	<i>80+ / pre_frail</i>	365	8	26.77	10	0.0028	10	0.257	-0.002

† Fewer risk groups than requested were usable, because the subgroup was small or its predicted probabilities were tied. Degrees of freedom follow the reduced count. A well-calibrated model gives a decile slope near 1 and a decile intercept near 0 (Kramer & Zimmerman 2007). 73 tests are reported here; no multiplicity adjustment has been applied. Adjust within an attribute if needed — do not pool the “Overall” rows with the subgroup rows.

Attribute	Metric	Value
sex	max_hl_gap	175.15
age_group	max_hl_gap	198.30
frailty	max_hl_gap	235.46
clinic	max_hl_gap	255.19
surgical_group	max_hl_gap	204.07
emergency	max_hl_gap	337.95
sex × age_group	max_hl_gap	146.14
sex × frailty	max_hl_gap	185.66
age_group × frailty	max_hl_gap	128.07

The gap is the largest pairwise difference in the statistic across subgroups. It inherits the sample-size dependence in full and should be read as a descriptive summary, not as evidence of differential calibration.

Observed versus expected events per decile

Deciles of predicted risk, of approximately equal size, with cut points snapped to tied predicted probabilities so that identical predictions always share a decile.

sex — Overall

Decile	n	Observed	Expected	Observed rate	Expected rate	Difference
1	1227	4	11.4	0.0033	0.0093	-7.4
2	833	4	18.9	0.0048	0.0227	-14.9
3	1322	10	46.0	0.0076	0.0348	-36.0
4	741	6	35.1	0.0081	0.0473	-29.1
5	1325	13	82.4	0.0098	0.0622	-69.4
6	1024	13	84.4	0.0127	0.0824	-71.4
7	1050	20	112.3	0.0190	0.1069	-92.3
8	1010	42	144.2	0.0416	0.1428	-102.2
9	1120	78	231.0	0.0696	0.2063	-153.0
10	1067	314	407.3	0.2943	0.3818	-93.3

sex — female

Decile	n	Observed	Expected	Observed rate	Expected rate	Difference
1	436	3	4.1	0.0069	0.0095	-1.1
2	278	1	6.3	0.0036	0.0227	-5.3
3	427	3	14.8	0.0070	0.0348	-11.8
4	389	3	19.3	0.0077	0.0496	-16.3
5	328	5	21.2	0.0152	0.0645	-16.2
6	343	4	28.2	0.0117	0.0823	-24.2
7	366	5	39.1	0.0137	0.1068	-34.1
8	393	19	57.9	0.0483	0.1473	-38.9
9	367	31	80.5	0.0845	0.2195	-49.5
10	362	116	145.1	0.3204	0.4008	-29.1

sex — male

Decile	n	Observed	Expected	Observed rate	Expected rate	Difference
1	791	1	7.2	0.0013	0.0091	-6.2
2	555	3	12.6	0.0054	0.0226	-9.6
3	895	7	31.2	0.0078	0.0348	-24.2
4	471	5	22.3	0.0106	0.0473	-17.3
5	878	6	54.7	0.0068	0.0623	-48.7
6	681	9	56.1	0.0132	0.0824	-47.1
7	684	15	73.2	0.0219	0.1070	-58.2
8	682	25	97.5	0.0367	0.1430	-72.5
9	679	47	137.5	0.0692	0.2025	-90.5
10	714	196	264.0	0.2745	0.3697	-68.0

age_group — Overall

Decile	n	Observed	Expected	Observed rate	Expected rate	Difference
1	1227	4	11.4	0.0033	0.0093	-7.4
2	833	4	18.9	0.0048	0.0227	-14.9
3	1322	10	46.0	0.0076	0.0348	-36.0
4	741	6	35.1	0.0081	0.0473	-29.1
5	1325	13	82.4	0.0098	0.0622	-69.4
6	1024	13	84.4	0.0127	0.0824	-71.4
7	1050	20	112.3	0.0190	0.1069	-92.3
8	1010	42	144.2	0.0416	0.1428	-102.2
9	1120	78	231.0	0.0696	0.2063	-153.0
10	1067	314	407.3	0.2943	0.3818	-93.3

age_group — 0-17

Decile	n	Observed	Expected	Observed rate	Expected rate	Difference
1	37	0	0.9	0.0000	0.0253	-0.9
2	36	1	1.7	0.0278	0.0476	-0.7
3	38	0	2.5	0.0000	0.0649	-2.5
4	32	2	2.6	0.0625	0.0803	-0.6
5	45	0	4.4	0.0000	0.0980	-4.4
6	33	1	3.9	0.0303	0.1185	-2.9
7	39	1	5.5	0.0256	0.1419	-4.5
8	35	2	6.2	0.0571	0.1773	-4.2
9	38	4	8.4	0.1053	0.2204	-4.4
10	34	10	11.8	0.2941	0.3469	-1.8

age_group — 18-44

Decile	n	Observed	Expected	Observed rate	Expected rate	Difference
1	82	0	0.5	0.0000	0.0063	-0.5
2	105	0	2.1	0.0000	0.0198	-2.1
3	85	0	2.8	0.0000	0.0331	-2.8
4	115	1	5.1	0.0087	0.0444	-4.1
5	86	0	4.9	0.0000	0.0574	-4.9
6	100	4	7.3	0.0400	0.0730	-3.3
7	86	2	8.1	0.0233	0.0945	-6.1
8	98	4	12.6	0.0408	0.1285	-8.6
9	92	5	18.0	0.0543	0.1957	-13.0
10	92	18	33.4	0.1957	0.3627	-15.4

age_group — 45-64

Decile	n	Observed	Expected	Observed rate	Expected rate	Difference
1	348	1	2.2	0.0029	0.0064	-1.2
2	473	2	9.6	0.0042	0.0202	-7.6
3	321	2	10.4	0.0062	0.0324	-8.4
4	404	2	18.1	0.0050	0.0448	-16.1
5	336	0	20.2	0.0000	0.0601	-20.2
6	372	4	28.5	0.0108	0.0767	-24.5
7	362	6	36.8	0.0166	0.1016	-30.8
8	370	16	50.9	0.0432	0.1376	-34.9
9	363	17	72.2	0.0468	0.1990	-55.2
10	366	97	137.9	0.2650	0.3768	-40.9

age_group — 65-79

Decile	n	Observed	Expected	Observed rate	Expected rate	Difference
1	524	2	4.9	0.0038	0.0093	-2.9
2	380	2	8.6	0.0053	0.0226	-6.6
3	413	3	13.4	0.0073	0.0324	-10.4
4	498	5	22.2	0.0100	0.0446	-17.2
5	565	8	35.1	0.0142	0.0621	-27.1
6	458	1	37.9	0.0022	0.0828	-36.9
7	449	10	47.9	0.0223	0.1067	-37.9
8	443	18	63.3	0.0406	0.1430	-45.3
9	493	34	101.9	0.0690	0.2066	-67.9
10	461	166	179.7	0.3601	0.3898	-13.7

age_group — 80+

Decile	n	Observed	Expected	Observed rate	Expected rate	Difference
1	98	0	1.3	0.0000	0.0132	-1.3
2	105	2	3.2	0.0190	0.0300	-1.2
3	109	1	4.9	0.0092	0.0453	-3.9
4	99	3	5.9	0.0303	0.0596	-2.9
5	106	2	8.1	0.0189	0.0768	-6.1
6	100	0	9.9	0.0000	0.0992	-9.9
7	93	3	11.8	0.0323	0.1274	-8.8
8	102	8	18.1	0.0784	0.1774	-10.1
9	97	14	24.1	0.1443	0.2481	-10.1
10	103	20	41.0	0.1942	0.3983	-21.0

frailty — Overall

Decile	n	Observed	Expected	Observed rate	Expected rate	Difference
1	1227	4	11.4	0.0033	0.0093	-7.4
2	833	4	18.9	0.0048	0.0227	-14.9
3	1322	10	46.0	0.0076	0.0348	-36.0
4	741	6	35.1	0.0081	0.0473	-29.1
5	1325	13	82.4	0.0098	0.0622	-69.4
6	1024	13	84.4	0.0127	0.0824	-71.4
7	1050	20	112.3	0.0190	0.1069	-92.3
8	1010	42	144.2	0.0416	0.1428	-102.2
9	1120	78	231.0	0.0696	0.2063	-153.0
10	1067	314	407.3	0.2943	0.3818	-93.3

frailty — frail

Decile	n	Observed	Expected	Observed rate	Expected rate	Difference
1	604	2	7.8	0.0033	0.0129	-5.8
2	688	7	20.8	0.0102	0.0302	-13.8
3	608	3	27.3	0.0049	0.0449	-24.3
4	577	5	34.6	0.0087	0.0600	-29.6
5	646	9	49.9	0.0139	0.0773	-40.9
6	595	12	59.2	0.0202	0.0995	-47.2
7	586	21	75.1	0.0358	0.1281	-54.1
8	597	28	102.0	0.0469	0.1708	-74.0
9	636	94	154.7	0.1478	0.2432	-60.7
10	602	219	259.7	0.3638	0.4314	-40.7

frailty — non_frail

Decile	n	Observed	Expected	Observed rate	Expected rate	Difference
1	94	0	0.3	0.0000	0.0030	-0.3
2	143	0	1.8	0.0000	0.0124	-1.8
3	61	0	1.2	0.0000	0.0200	-1.2
4	106	0	2.9	0.0000	0.0273	-2.9
5	110	0	4.1	0.0000	0.0373	-4.1
6	96	1	4.5	0.0104	0.0469	-3.5
7	113	0	6.7	0.0000	0.0596	-6.7
8	97	0	7.7	0.0000	0.0796	-7.7
9	98	2	11.1	0.0204	0.1133	-9.1
10	104	3	21.9	0.0288	0.2108	-18.9

frailty — pre_frail

Decile	n	Observed	Expected	Observed rate	Expected rate	Difference
1	382	1	2.3	0.0026	0.0062	-1.3
2	333	1	5.8	0.0030	0.0173	-4.8
3	354	2	9.7	0.0056	0.0275	-7.7
4	293	4	11.0	0.0137	0.0375	-7.0
5	367	4	18.2	0.0109	0.0496	-14.2
6	410	5	27.6	0.0122	0.0672	-22.6
7	372	5	33.1	0.0134	0.0891	-28.1
8	330	10	39.3	0.0303	0.1192	-29.3
9	368	16	63.2	0.0435	0.1717	-47.2
10	349	50	109.3	0.1433	0.3133	-59.3

clinic — gynecology

Decile	n	Observed	Expected	Observed rate	Expected rate	Difference
1	46	0	0.5	0.0000	0.0100	-0.5
2	46	0	1.0	0.0000	0.0227	-1.0
3	45	0	1.5	0.0000	0.0331	-1.5
4	39	2	1.7	0.0513	0.0429	+0.3
5	52	2	2.9	0.0385	0.0555	-0.9
6	41	1	3.0	0.0244	0.0741	-2.0
7	48	1	4.5	0.0208	0.0936	-3.5
8	45	3	5.4	0.0667	0.1208	-2.4
9	45	3	7.9	0.0667	0.1762	-4.9
10	45	12	14.6	0.2667	0.3251	-2.6

clinic — Overall

Decile	n	Observed	Expected	Observed rate	Expected rate	Difference
1	1227	4	11.4	0.0033	0.0093	-7.4
2	833	4	18.9	0.0048	0.0227	-14.9
3	1322	10	46.0	0.0076	0.0348	-36.0
4	741	6	35.1	0.0081	0.0473	-29.1
5	1325	13	82.4	0.0098	0.0622	-69.4
6	1024	13	84.4	0.0127	0.0824	-71.4
7	1050	20	112.3	0.0190	0.1069	-92.3
8	1010	42	144.2	0.0416	0.1428	-102.2
9	1120	78	231.0	0.0696	0.2063	-153.0
10	1067	314	407.3	0.2943	0.3818	-93.3

clinic — neuro

Decile	n	Observed	Expected	Observed rate	Expected rate	Difference
1	128	1	0.8	0.0078	0.0059	+0.2
2	116	0	2.0	0.0000	0.0172	-2.0
3	63	0	1.6	0.0000	0.0250	-1.6
4	165	0	5.7	0.0000	0.0347	-5.7
5	91	0	4.3	0.0000	0.0474	-4.3
6	94	0	5.6	0.0000	0.0593	-5.6
7	135	4	10.4	0.0296	0.0771	-6.4
8	102	1	10.2	0.0098	0.1001	-9.2
9	115	4	15.9	0.0348	0.1385	-11.9
10	108	16	27.8	0.1481	0.2574	-11.8

clinic — ORL

Decile	n	Observed	Expected	Observed rate	Expected rate	Difference
1	24	0	0.2	0.0000	0.0100	-0.2
2	26	0	0.8	0.0000	0.0325	-0.8
3	23	0	1.4	0.0000	0.0630	-1.4
4	20	0	1.8	0.0000	0.0885	-1.8
5	23	2	2.7	0.0870	0.1170	-0.7
6	24	1	3.4	0.0417	0.1425	-2.4
7	23	0	4.2	0.0000	0.1830	-4.2
8	24	2	5.7	0.0833	0.2369	-3.7
9	23	6	7.3	0.2609	0.3193	-1.3
10	23	4	10.3	0.1739	0.4463	-6.3

clinic — ortho

Decile	n	Observed	Expected	Observed rate	Expected rate	Difference
1	158	0	0.9	0.0000	0.0058	-0.9
2	199	1	4.0	0.0050	0.0201	-3.0
3	169	1	5.9	0.0059	0.0349	-4.9
4	167	2	8.4	0.0120	0.0501	-6.4
5	161	2	10.4	0.0124	0.0646	-8.4
6	146	1	12.0	0.0068	0.0822	-11.0
7	183	2	19.8	0.0109	0.1084	-17.8
8	172	12	26.1	0.0698	0.1520	-14.1
9	164	20	35.3	0.1220	0.2151	-15.3
10	171	35	63.7	0.2047	0.3726	-28.7

clinic — pediatric

Decile	n	Observed	Expected	Observed rate	Expected rate	Difference
1	7	0	0.2	0.0000	0.0229	-0.2
2	7	1	0.4	0.1429	0.0507	+0.6
3	7	0	0.5	0.0000	0.0729	-0.5
4	8	0	0.7	0.0000	0.0900	-0.7
5	5	0	0.5	0.0000	0.1070	-0.5
6	6	0	0.7	0.0000	0.1233	-0.7
7	7	0	1.0	0.0000	0.1443	-1.0
8	7	1	1.4	0.1429	0.1936	-0.4
9	5	0	1.2	0.0000	0.2500	-1.2
10	6	1	2.0	0.1667	0.3375	-1.0

clinic — vascular

Decile	n	Observed	Expected	Observed rate	Expected rate	Difference
1	456	2	4.4	0.0044	0.0096	-2.4
2	602	4	15.3	0.0066	0.0254	-11.3
3	423	5	15.8	0.0118	0.0374	-10.8
4	536	3	26.7	0.0056	0.0498	-23.7
5	479	3	31.1	0.0063	0.0649	-28.1
6	371	2	29.7	0.0054	0.0801	-27.7
7	563	9	57.1	0.0160	0.1015	-48.1
8	455	14	61.6	0.0308	0.1354	-47.6
9	483	22	91.2	0.0455	0.1888	-69.2
10	484	120	161.3	0.2479	0.3332	-41.3

clinic — thoracic

Decile	n	Observed	Expected	Observed rate	Expected rate	Difference
1	57	0	0.2	0.0000	0.0028	-0.2
2	101	0	1.3	0.0000	0.0125	-1.3
3	43	0	0.9	0.0000	0.0200	-0.9
4	85	0	2.3	0.0000	0.0276	-2.3
5	80	0	3.0	0.0000	0.0378	-3.0
6	66	0	3.3	0.0000	0.0493	-3.3
7	77	0	5.1	0.0000	0.0660	-5.1
8	75	1	7.1	0.0133	0.0951	-6.1
9	68	3	9.8	0.0441	0.1439	-6.8
10	71	10	21.6	0.1408	0.3042	-11.6

clinic — visceral

Decile	n	Observed	Expected	Observed rate	Expected rate	Difference
1	116	0	1.4	0.0000	0.0117	-1.4
2	125	1	4.0	0.0080	0.0324	-3.0
3	146	2	8.0	0.0137	0.0550	-6.0
4	120	1	9.5	0.0083	0.0793	-8.5
5	109	4	11.6	0.0367	0.1068	-7.6
6	129	6	18.8	0.0465	0.1460	-12.8
7	125	7	25.3	0.0560	0.2024	-18.3
8	126	25	34.2	0.1984	0.2718	-9.2
9	120	25	44.6	0.2083	0.3714	-19.6
10	122	72	69.4	0.5902	0.5685	+2.6

clinic — urology

Decile	n	Observed	Expected	Observed rate	Expected rate	Difference
1	37	1	0.4	0.0270	0.0114	+0.6
2	23	0	0.6	0.0000	0.0274	-0.6
3	43	1	1.7	0.0233	0.0391	-0.7
4	37	0	2.1	0.0000	0.0554	-2.1
5	35	2	2.4	0.0571	0.0693	-0.4
6	36	1	3.1	0.0278	0.0869	-2.1
7	35	0	3.9	0.0000	0.1126	-3.9
8	34	2	4.8	0.0588	0.1415	-2.8
9	35	1	6.6	0.0286	0.1894	-5.6
10	34	11	11.1	0.3235	0.3265	-0.1

surgical_group — Overall

Decile	n	Observed	Expected	Observed rate	Expected rate	Difference
1	1227	4	11.4	0.0033	0.0093	-7.4
2	833	4	18.9	0.0048	0.0227	-14.9
3	1322	10	46.0	0.0076	0.0348	-36.0
4	741	6	35.1	0.0081	0.0473	-29.1
5	1325	13	82.4	0.0098	0.0622	-69.4
6	1024	13	84.4	0.0127	0.0824	-71.4
7	1050	20	112.3	0.0190	0.1069	-92.3
8	1010	42	144.2	0.0416	0.1428	-102.2
9	1120	78	231.0	0.0696	0.2063	-153.0
10	1067	314	407.3	0.2943	0.3818	-93.3

surgical_group — chopcluster_1

Decile	n	Observed	Expected	Observed rate	Expected rate	Difference
1	107	1	0.6	0.0093	0.0059	+0.4
2	92	0	1.6	0.0000	0.0171	-1.6
3	92	0	2.5	0.0000	0.0274	-2.5
4	86	0	3.2	0.0000	0.0373	-3.2
5	76	0	3.6	0.0000	0.0474	-3.6
6	77	0	4.6	0.0000	0.0593	-4.6
7	102	3	7.8	0.0294	0.0766	-4.8
8	88	1	8.8	0.0114	0.1006	-7.8
9	94	2	13.3	0.0213	0.1417	-11.3
10	89	15	25.2	0.1685	0.2833	-10.2

surgical_group — chopcluster_10

Decile	n	Observed	Expected	Observed rate	Expected rate	Difference
1	34	1	0.5	0.0294	0.0151	+0.5
2	32	1	1.1	0.0312	0.0345	-0.1
3	36	1	1.8	0.0278	0.0504	-0.8
4	42	2	2.8	0.0476	0.0673	-0.8
5	32	1	2.7	0.0312	0.0859	-1.7
6	32	0	3.4	0.0000	0.1052	-3.4
7	38	1	5.0	0.0263	0.1317	-4.0
8	34	3	5.7	0.0882	0.1663	-2.7
9	33	3	7.1	0.0909	0.2162	-4.1
10	33	15	11.9	0.4545	0.3594	+3.1

surgical_group — chopcluster_11

Decile	n	Observed	Expected	Observed rate	Expected rate	Difference
1	29	0	0.2	0.0000	0.0078	-0.2
2	30	0	0.5	0.0000	0.0180	-0.5
3	33	0	0.9	0.0000	0.0271	-0.9
4	30	0	1.1	0.0000	0.0370	-1.1
5	34	2	1.7	0.0588	0.0487	+0.3
6	22	1	1.4	0.0455	0.0623	-0.4
7	37	1	2.9	0.0270	0.0793	-1.9
8	32	1	3.2	0.0312	0.0997	-2.2
9	26	1	3.5	0.0385	0.1362	-2.5
10	30	3	8.6	0.1000	0.2860	-5.6

surgical_group — chopcluster_12

Decile	n	Observed	Expected	Observed rate	Expected rate	Difference
1	159	0	0.9	0.0000	0.0059	-0.9
2	197	1	3.9	0.0051	0.0201	-2.9
3	108	0	3.5	0.0000	0.0322	-3.5
4	166	2	7.4	0.0120	0.0446	-5.4
5	168	0	10.1	0.0000	0.0600	-10.1
6	169	2	13.1	0.0118	0.0774	-11.1
7	172	3	17.5	0.0174	0.1016	-14.5
8	166	10	23.5	0.0602	0.1414	-13.5
9	153	12	30.6	0.0784	0.2003	-18.6
10	165	32	58.2	0.1939	0.3530	-26.2

surgical_group — chopcluster_13

Decile	n	Observed	Expected	Observed rate	Expected rate	Difference
1	24	1	1.2	0.0417	0.0500	-0.2
2	20	2	1.6	0.1000	0.0808	+0.4
3	22	1	2.3	0.0455	0.1066	-1.3
4	21	1	2.8	0.0476	0.1312	-1.8
5	22	2	3.5	0.0909	0.1607	-1.5
6	19	2	3.8	0.1053	0.1995	-1.8
7	22	6	5.2	0.2727	0.2343	+0.8
8	21	7	6.4	0.3333	0.3048	+0.6
9	21	7	8.2	0.3333	0.3888	-1.2
10	21	14	11.5	0.6667	0.5464	+2.5

surgical_group — chopcluster_3

Decile	n	Observed	Expected	Observed rate	Expected rate	Difference
1	58	0	0.2	0.0000	0.0030	-0.2
2	109	0	1.4	0.0000	0.0126	-1.4
3	91	0	2.0	0.0000	0.0224	-2.0
4	81	0	2.6	0.0000	0.0322	-2.6
5	78	0	3.3	0.0000	0.0421	-3.3
6	78	0	4.3	0.0000	0.0557	-4.3
7	73	1	5.4	0.0137	0.0743	-4.4
8	76	0	8.0	0.0000	0.1057	-8.0
9	84	4	14.1	0.0476	0.1684	-10.1
10	79	17	28.0	0.2152	0.3541	-11.0

surgical_group — chopcluster_4

Decile	n	Observed	Expected	Observed rate	Expected rate	Difference
1	338	2	3.4	0.0059	0.0100	-1.4
2	481	4	12.3	0.0083	0.0255	-8.3
3	351	4	13.1	0.0114	0.0373	-9.1
4	302	1	14.3	0.0033	0.0475	-13.3
5	409	3	24.5	0.0073	0.0600	-21.5
6	426	4	32.8	0.0094	0.0771	-28.8
7	302	2	29.2	0.0066	0.0967	-27.2
8	396	10	49.0	0.0253	0.1237	-39.0
9	380	12	65.8	0.0316	0.1731	-53.8
10	379	76	116.0	0.2005	0.3061	-40.0

surgical_group — chopcluster_5

Decile	n	Observed	Expected	Observed rate	Expected rate	Difference
1	67	0	0.4	0.0000	0.0062	-0.4
2	96	0	1.9	0.0000	0.0201	-1.9
3	65	1	2.1	0.0154	0.0324	-1.1
4	72	0	3.2	0.0000	0.0448	-3.2
5	80	0	5.0	0.0000	0.0628	-5.0
6	77	1	6.3	0.0130	0.0822	-5.3
7	80	1	8.6	0.0125	0.1078	-7.6
8	82	3	12.0	0.0366	0.1465	-9.0
9	76	4	15.8	0.0526	0.2076	-11.8
10	74	20	27.2	0.2703	0.3673	-7.2

surgical_group — chopcluster_6

Decile	n	Observed	Expected	Observed rate	Expected rate	Difference
1	55	0	1.2	0.0000	0.0215	-1.2
2	50	1	2.0	0.0200	0.0406	-1.0
3	52	0	3.0	0.0000	0.0573	-3.0
4	59	1	4.7	0.0169	0.0800	-3.7
5	49	3	5.4	0.0612	0.1108	-2.4
6	50	2	7.3	0.0400	0.1467	-5.3
7	58	3	11.5	0.0517	0.1982	-8.5
8	50	8	12.9	0.1600	0.2570	-4.9
9	54	16	19.5	0.2963	0.3618	-3.5
10	53	33	31.3	0.6226	0.5904	+1.7

surgical_group — chopcluster_7

Decile	n	Observed	Expected	Observed rate	Expected rate	Difference
1	32	0	1.1	0.0000	0.0344	-1.1
2	30	0	2.0	0.0000	0.0657	-2.0
3	31	0	2.9	0.0000	0.0929	-2.9
4	27	0	3.2	0.0000	0.1204	-3.2
5	31	1	4.8	0.0323	0.1548	-3.8
6	30	4	6.1	0.1333	0.2025	-2.1
7	28	4	7.0	0.1429	0.2507	-3.0
8	31	6	9.4	0.1935	0.3029	-3.4
9	29	5	10.9	0.1724	0.3755	-5.9
10	29	16	16.4	0.5517	0.5657	-0.4

surgical_group — chopcluster_8

Decile	n	Observed	Expected	Observed rate	Expected rate	Difference
1	30	0	0.4	0.0000	0.0127	-0.4
2	34	1	1.1	0.0294	0.0334	-0.1
3	30	2	1.5	0.0667	0.0498	+0.5
4	28	0	1.8	0.0000	0.0646	-1.8
5	37	0	3.2	0.0000	0.0857	-3.2
6	26	2	3.0	0.0769	0.1156	-1.0
7	33	2	5.2	0.0606	0.1562	-3.2
8	29	0	6.3	0.0000	0.2164	-6.3
9	29	6	8.8	0.2069	0.3021	-2.8
10	30	10	14.9	0.3333	0.4953	-4.9

surgical_group — none

Decile	n	Observed	Expected	Observed rate	Expected rate	Difference
1	98	0	0.6	0.0000	0.0061	-0.6
2	71	0	1.2	0.0000	0.0175	-1.2
3	75	0	2.1	0.0000	0.0277	-2.1
4	105	0	4.3	0.0000	0.0408	-4.3
5	88	1	5.5	0.0114	0.0622	-4.5
6	72	0	6.1	0.0000	0.0849	-6.1
7	96	5	11.2	0.0521	0.1168	-6.2
8	85	4	13.8	0.0471	0.1625	-9.8
9	83	10	19.3	0.1205	0.2326	-9.3
10	84	24	35.1	0.2857	0.4176	-11.1

emergency — Overall

Decile	n	Observed	Expected	Observed rate	Expected rate	Difference
1	1227	4	11.4	0.0033	0.0093	-7.4
2	833	4	18.9	0.0048	0.0227	-14.9
3	1322	10	46.0	0.0076	0.0348	-36.0
4	741	6	35.1	0.0081	0.0473	-29.1
5	1325	13	82.4	0.0098	0.0622	-69.4
6	1024	13	84.4	0.0127	0.0824	-71.4
7	1050	20	112.3	0.0190	0.1069	-92.3
8	1010	42	144.2	0.0416	0.1428	-102.2
9	1120	78	231.0	0.0696	0.2063	-153.0
10	1067	314	407.3	0.2943	0.3818	-93.3

sex × age_group — Overall

Decile	n	Observed	Expected	Observed rate	Expected rate	Difference
1	1227	4	11.4	0.0033	0.0093	-7.4
2	833	4	18.9	0.0048	0.0227	-14.9
3	1322	10	46.0	0.0076	0.0348	-36.0
4	741	6	35.1	0.0081	0.0473	-29.1
5	1325	13	82.4	0.0098	0.0622	-69.4
6	1024	13	84.4	0.0127	0.0824	-71.4
7	1050	20	112.3	0.0190	0.1069	-92.3
8	1010	42	144.2	0.0416	0.1428	-102.2
9	1120	78	231.0	0.0696	0.2063	-153.0
10	1067	314	407.3	0.2943	0.3818	-93.3

emergency — elective

Decile	n	Observed	Expected	Observed rate	Expected rate	Difference
1	819	2	5.3	0.0024	0.0064	-3.3
2	1219	6	24.5	0.0049	0.0201	-18.5
3	890	5	28.9	0.0056	0.0325	-23.9
4	772	7	32.8	0.0091	0.0424	-25.8
5	976	9	53.7	0.0092	0.0550	-44.7
6	813	10	56.6	0.0123	0.0696	-46.6
7	993	13	88.4	0.0131	0.0890	-75.4
8	884	21	105.2	0.0238	0.1190	-84.2
9	952	44	163.2	0.0462	0.1714	-119.2
10	932	199	301.3	0.2135	0.3233	-102.3

sex × age_group — female / 0-17

Decile	n	Observed	Expected	Observed rate	Expected rate	Difference
1	20	0	0.5	0.0000	0.0273	-0.5
2	15	0	0.7	0.0000	0.0487	-0.7
3	20	0	1.4	0.0000	0.0700	-1.4
4	18	0	1.6	0.0000	0.0894	-1.6
5	16	0	1.7	0.0000	0.1047	-1.7
6	18	0	2.4	0.0000	0.1317	-2.4
7	17	2	2.8	0.1176	0.1638	-0.8
8	17	0	3.3	0.0000	0.1935	-3.3
9	17	5	4.0	0.2941	0.2374	+1.0
10	17	6	6.1	0.3529	0.3600	-0.1

emergency — emergency

Decile	n	Observed	Expected	Observed rate	Expected rate	Difference
1	148	1	6.5	0.0068	0.0438	-5.5
2	132	1	10.0	0.0076	0.0759	-9.0
3	148	1	14.8	0.0068	0.0998	-13.8
4	167	11	21.0	0.0659	0.1259	-10.0
5	153	7	24.1	0.0458	0.1577	-17.1
6	140	7	26.6	0.0500	0.1901	-19.6
7	137	10	30.5	0.0730	0.2226	-20.5
8	156	25	43.7	0.1603	0.2802	-18.7
9	144	37	53.9	0.2569	0.3744	-16.9
10	144	88	81.9	0.6111	0.5689	+6.1

sex × age_group — female / 18-44

Decile	n	Observed	Expected	Observed rate	Expected rate	Difference
1	41	0	0.4	0.0000	0.0091	-0.4
2	35	0	0.9	0.0000	0.0254	-0.9
3	43	0	1.6	0.0000	0.0374	-1.6
4	51	0	2.6	0.0000	0.0501	-2.6
5	37	0	2.3	0.0000	0.0615	-2.3
6	40	2	3.1	0.0500	0.0773	-1.1
7	38	0	3.7	0.0000	0.0968	-3.7
8	44	1	5.6	0.0227	0.1275	-4.6
9	40	2	8.1	0.0500	0.2019	-6.1
10	39	8	14.6	0.2051	0.3755	-6.6

sex × age_group — female / 45-64

Decile	n	Observed	Expected	Observed rate	Expected rate	Difference
1	122	1	0.8	0.0082	0.0064	+0.2
2	111	1	1.9	0.0090	0.0173	-0.9
3	107	1	2.9	0.0093	0.0273	-1.9
4	138	2	5.5	0.0145	0.0400	-3.5
5	101	0	5.5	0.0000	0.0550	-5.5
6	131	2	9.5	0.0153	0.0726	-7.5
7	116	1	11.3	0.0086	0.0970	-10.3
8	122	5	16.3	0.0410	0.1338	-11.3
9	113	3	22.5	0.0265	0.1987	-19.5
10	118	36	46.7	0.3051	0.3954	-10.7

sex × age_group — female / 65-79

Decile	n	Observed	Expected	Observed rate	Expected rate	Difference
1	170	1	1.6	0.0059	0.0094	-0.6
2	115	0	2.6	0.0000	0.0226	-2.6
3	176	0	6.1	0.0000	0.0344	-6.1
4	149	1	7.4	0.0067	0.0497	-6.4
5	132	4	8.6	0.0303	0.0648	-4.6
6	130	0	10.8	0.0000	0.0832	-10.8
7	159	4	17.3	0.0252	0.1089	-13.3
8	145	11	22.2	0.0759	0.1531	-11.2
9	147	14	32.8	0.0952	0.2235	-18.8
10	147	58	59.8	0.3946	0.4070	-1.8

sex × age_group — female / 80+

Decile	n	Observed	Expected	Observed rate	Expected rate	Difference
1	48	0	0.6	0.0000	0.0128	-0.6
2	49	1	1.4	0.0204	0.0294	-0.4
3	32	0	1.4	0.0000	0.0428	-1.4
4	58	2	3.2	0.0345	0.0544	-1.2
5	40	0	2.8	0.0000	0.0704	-2.8
6	51	0	4.8	0.0000	0.0949	-4.8
7	43	0	5.3	0.0000	0.1222	-5.3
8	46	1	7.7	0.0217	0.1671	-6.7
9	43	8	11.0	0.1860	0.2560	-3.0
10	47	7	18.9	0.1489	0.4030	-11.9

sex × age_group — male / 0-17

Decile	n	Observed	Expected	Observed rate	Expected rate	Difference
1	21	0	0.6	0.0000	0.0262	-0.6
2	17	1	0.8	0.0588	0.0485	+0.2
3	18	0	1.1	0.0000	0.0622	-1.1
4	26	2	2.0	0.0769	0.0771	-0.0
5	12	0	1.1	0.0000	0.0950	-1.1
6	26	1	2.9	0.0385	0.1123	-1.9
7	16	0	2.1	0.0000	0.1313	-2.1
8	18	1	2.8	0.0556	0.1581	-1.8
9	20	0	4.0	0.0000	0.2010	-4.0
10	18	3	5.8	0.1667	0.3239	-2.8

sex × age_group — male / 18-44

Decile	n	Observed	Expected	Observed rate	Expected rate	Difference
1	53	0	0.3	0.0000	0.0060	-0.3
2	49	0	0.8	0.0000	0.0172	-0.8
3	72	0	2.2	0.0000	0.0307	-2.2
4	46	0	2.0	0.0000	0.0427	-2.0
5	42	1	2.2	0.0238	0.0529	-1.2
6	55	2	3.7	0.0364	0.0675	-1.7
7	57	2	5.1	0.0351	0.0900	-3.1
8	54	3	7.0	0.0556	0.1294	-4.0
9	51	3	9.7	0.0588	0.1902	-6.7
10	54	10	19.0	0.1852	0.3510	-9.0

sex × age_group — male / 45-64

Decile	n	Observed	Expected	Observed rate	Expected rate	Difference
1	226	0	1.4	0.0000	0.0064	-1.4
2	304	0	6.2	0.0000	0.0203	-6.2
3	224	2	7.3	0.0089	0.0324	-5.3
4	280	0	12.6	0.0000	0.0449	-12.6
5	237	0	14.3	0.0000	0.0602	-14.3
6	247	2	18.9	0.0081	0.0765	-16.9
7	260	5	26.3	0.0192	0.1013	-21.3
8	260	11	36.0	0.0423	0.1383	-25.0
9	241	13	47.6	0.0539	0.1974	-34.6
10	257	62	93.5	0.2412	0.3636	-31.5

sex × age_group — male / 65-79

Decile	n	Observed	Expected	Observed rate	Expected rate	Difference
1	354	1	3.3	0.0028	0.0093	-2.3
2	265	2	6.0	0.0075	0.0225	-4.0
3	283	3	9.2	0.0106	0.0324	-6.2
4	349	4	15.5	0.0115	0.0443	-11.5
5	387	4	24.0	0.0103	0.0621	-20.0
6	328	1	27.1	0.0030	0.0827	-26.1
7	268	5	28.1	0.0187	0.1048	-23.1
8	334	10	46.3	0.0299	0.1387	-36.3
9	324	19	64.6	0.0586	0.1994	-45.6
10	322	107	121.6	0.3323	0.3777	-14.6

sex × frailty — female / frail

Decile	n	Observed	Expected	Observed rate	Expected rate	Difference
1	165	1	2.7	0.0061	0.0162	-1.7
2	162	1	5.6	0.0062	0.0348	-4.6
3	170	1	8.5	0.0059	0.0500	-7.5
4	180	5	11.9	0.0278	0.0663	-6.9
5	146	0	12.7	0.0000	0.0868	-12.7
6	194	3	21.6	0.0155	0.1116	-18.6
7	142	5	20.7	0.0352	0.1461	-15.7
8	175	6	33.9	0.0343	0.1935	-27.9
9	167	40	47.0	0.2395	0.2812	-7.0
10	165	67	78.2	0.4061	0.4736	-11.2

sex × age_group — male / 80+

Decile	n	Observed	Expected	Observed rate	Expected rate	Difference
1	50	0	0.7	0.0000	0.0135	-0.7
2	56	1	1.7	0.0179	0.0305	-0.7
3	69	1	3.3	0.0145	0.0471	-2.3
4	59	1	3.8	0.0169	0.0649	-2.8
5	44	2	3.5	0.0455	0.0803	-1.5
6	54	0	5.3	0.0000	0.0988	-5.3
7	60	3	7.9	0.0500	0.1323	-4.9
8	51	7	9.4	0.1373	0.1845	-2.4
9	56	7	13.6	0.1250	0.2421	-6.6
10	56	12	22.0	0.2143	0.3931	-10.0

sex × frailty — female / non_frail

Decile	n	Observed	Expected	Observed rate	Expected rate	Difference
1	38	0	0.1	0.0000	0.0028	-0.1
2	69	0	0.9	0.0000	0.0125	-0.9
3	58	0	1.3	0.0000	0.0224	-1.3
4	52	0	1.7	0.0000	0.0329	-1.7
5	27	0	1.1	0.0000	0.0400	-1.1
6	73	0	3.6	0.0000	0.0488	-3.6
7	48	0	3.0	0.0000	0.0633	-3.0
8	55	0	4.7	0.0000	0.0862	-4.7
9	50	0	6.0	0.0000	0.1193	-6.0
10	52	1	11.9	0.0192	0.2287	-10.9

sex × frailty — Overall

Decile	n	Observed	Expected	Observed rate	Expected rate	Difference
1	1227	4	11.4	0.0033	0.0093	-7.4
2	833	4	18.9	0.0048	0.0227	-14.9
3	1322	10	46.0	0.0076	0.0348	-36.0
4	741	6	35.1	0.0081	0.0473	-29.1
5	1325	13	82.4	0.0098	0.0622	-69.4
6	1024	13	84.4	0.0127	0.0824	-71.4
7	1050	20	112.3	0.0190	0.1069	-92.3
8	1010	42	144.2	0.0416	0.1428	-102.2
9	1120	78	231.0	0.0696	0.2063	-153.0
10	1067	314	407.3	0.2943	0.3818	-93.3

sex × frailty — female / pre_frail

Decile	n	Observed	Expected	Observed rate	Expected rate	Difference
1	157	1	1.0	0.0064	0.0063	+0.0
2	144	1	2.5	0.0069	0.0170	-1.5
3	157	1	4.3	0.0064	0.0274	-3.3
4	167	3	6.7	0.0180	0.0399	-3.7
5	138	1	7.6	0.0072	0.0550	-6.6
6	129	2	9.1	0.0155	0.0703	-7.1
7	156	2	13.8	0.0128	0.0886	-11.8
8	153	9	18.6	0.0588	0.1218	-9.6
9	154	12	27.9	0.0779	0.1812	-15.9
10	146	28	48.2	0.1918	0.3299	-20.2

sex × frailty — male / frail

Decile	n	Observed	Expected	Observed rate	Expected rate	Difference
1	480	1	6.1	0.0021	0.0128	-5.1
2	350	4	9.6	0.0114	0.0276	-5.6
3	492	5	19.5	0.0102	0.0396	-14.5
4	409	3	22.5	0.0073	0.0550	-19.5
5	504	3	36.1	0.0060	0.0717	-33.1
6	463	8	43.2	0.0173	0.0933	-35.2
7	423	16	51.3	0.0378	0.1214	-35.3
8	451	23	72.8	0.0510	0.1615	-49.8
9	457	56	104.7	0.1225	0.2290	-48.7
10	444	152	182.2	0.3423	0.4105	-30.2

age_group × frailty — Overall

Decile	n	Observed	Expected	Observed rate	Expected rate	Difference
1	1227	4	11.4	0.0033	0.0093	-7.4
2	833	4	18.9	0.0048	0.0227	-14.9
3	1322	10	46.0	0.0076	0.0348	-36.0
4	741	6	35.1	0.0081	0.0473	-29.1
5	1325	13	82.4	0.0098	0.0622	-69.4
6	1024	13	84.4	0.0127	0.0824	-71.4
7	1050	20	112.3	0.0190	0.1069	-92.3
8	1010	42	144.2	0.0416	0.1428	-102.2
9	1120	78	231.0	0.0696	0.2063	-153.0
10	1067	314	407.3	0.2943	0.3818	-93.3

sex × frailty — male / non_frail

Decile	n	Observed	Expected	Observed rate	Expected rate	Difference
1	56	0	0.2	0.0000	0.0032	-0.2
2	40	0	0.4	0.0000	0.0100	-0.4
3	65	0	1.1	0.0000	0.0174	-1.1
4	29	0	0.7	0.0000	0.0250	-0.7
5	57	0	1.9	0.0000	0.0326	-1.9
6	45	0	1.9	0.0000	0.0424	-1.9
7	57	1	3.1	0.0175	0.0544	-2.1
8	49	0	3.5	0.0000	0.0712	-3.5
9	50	2	5.2	0.0400	0.1045	-3.2
10	52	2	10.0	0.0385	0.1920	-8.0

age_group × frailty — 0-17 / frail

Decile	n	Observed	Expected	Observed rate	Expected rate	Difference
1	23	0	0.9	0.0000	0.0370	-0.9
2	20	0	1.3	0.0000	0.0638	-1.3
3	18	2	1.5	0.1111	0.0831	+0.5
4	23	0	2.3	0.0000	0.1020	-2.3
5	26	1	3.3	0.0385	0.1256	-2.3
6	17	1	2.5	0.0588	0.1488	-1.5
7	22	1	4.0	0.0455	0.1798	-3.0
8	18	2	3.8	0.1111	0.2106	-1.8
9	23	5	5.8	0.2174	0.2502	-0.8
10	21	7	8.2	0.3333	0.3893	-1.2

sex × frailty — male / pre_frail

Decile	n	Observed	Expected	Observed rate	Expected rate	Difference
1	225	0	1.4	0.0000	0.0061	-1.4
2	189	0	3.3	0.0000	0.0175	-3.3
3	197	1	5.4	0.0051	0.0277	-4.4
4	181	2	6.8	0.0110	0.0376	-4.8
5	221	2	11.0	0.0090	0.0496	-9.0
6	234	3	15.7	0.0128	0.0670	-12.7
7	178	2	15.5	0.0112	0.0872	-13.5
8	232	3	26.9	0.0129	0.1159	-23.9
9	196	4	32.9	0.0204	0.1678	-28.9
10	204	21	61.2	0.1029	0.3000	-40.2

age_group × frailty — 0-17 / non_frail

Decile	n	Observed	Expected	Observed rate	Expected rate	Difference
1	4	0	0.1	0.0000	0.0163	-0.1
2	5	0	0.1	0.0000	0.0270	-0.1
3	6	0	0.2	0.0000	0.0375	-0.2
4	6	0	0.3	0.0000	0.0525	-0.3
5	4	0	0.3	0.0000	0.0675	-0.3
6	3	0	0.2	0.0000	0.0767	-0.2
7	4	0	0.3	0.0000	0.0862	-0.3
8	6	0	0.6	0.0000	0.1017	-0.6
9	3	0	0.4	0.0000	0.1217	-0.4
10	4	0	0.7	0.0000	0.1713	-0.7

age_group × frailty — 0-17 / pre_frail

Decile	n	Observed	Expected	Observed rate	Expected rate	Difference
1	14	0	0.3	0.0000	0.0232	-0.3
2	9	1	0.4	0.1111	0.0450	+0.6
3	11	0	0.6	0.0000	0.0582	-0.6
4	9	0	0.6	0.0000	0.0678	-0.6
5	13	0	1.0	0.0000	0.0800	-1.0
6	9	0	0.8	0.0000	0.0928	-0.8
7	12	0	1.3	0.0000	0.1121	-1.3
8	12	0	1.6	0.0000	0.1321	-1.6
9	12	1	2.0	0.0833	0.1696	-1.0
10	10	0	2.4	0.0000	0.2415	-2.4

age_group × frailty — 18-44 / pre_frail

Decile	n	Observed	Expected	Observed rate	Expected rate	Difference
1	28	0	0.2	0.0000	0.0059	-0.2
2	32	0	0.6	0.0000	0.0189	-0.6
3	28	0	0.9	0.0000	0.0325	-0.9
4	44	1	2.1	0.0227	0.0480	-1.1
5	26	1	1.6	0.0385	0.0631	-0.6
6	28	2	2.2	0.0714	0.0770	-0.2
7	33	0	3.1	0.0000	0.0944	-3.1
8	30	2	3.6	0.0667	0.1212	-1.6
9	31	2	5.3	0.0645	0.1697	-3.3
10	30	3	8.9	0.1000	0.2958	-5.9

age_group × frailty — 18-44 / frail

Decile	n	Observed	Expected	Observed rate	Expected rate	Difference
1	33	0	0.7	0.0000	0.0198	-0.7
2	24	0	0.9	0.0000	0.0377	-0.9
3	30	0	1.5	0.0000	0.0498	-1.5
4	32	1	2.1	0.0312	0.0669	-1.1
5	33	1	3.0	0.0303	0.0898	-2.0
6	25	0	2.9	0.0000	0.1172	-2.9
7	33	2	5.0	0.0606	0.1523	-3.0
8	30	2	6.4	0.0667	0.2133	-4.4
9	30	5	9.2	0.1667	0.3078	-4.2
10	30	10	14.1	0.3333	0.4685	-4.1

age_group × frailty — 45-64 / frail

Decile	n	Observed	Expected	Observed rate	Expected rate	Difference
1	174	1	1.7	0.0057	0.0100	-0.7
2	202	1	5.2	0.0050	0.0255	-4.2
3	211	0	8.3	0.0000	0.0394	-8.3
4	185	0	10.1	0.0000	0.0547	-10.1
5	232	2	16.6	0.0086	0.0715	-14.6
6	182	1	17.1	0.0055	0.0938	-16.1
7	190	8	23.1	0.0421	0.1218	-15.1
8	206	11	33.7	0.0534	0.1636	-22.7
9	194	19	46.2	0.0979	0.2379	-27.2
10	191	68	84.2	0.3560	0.4410	-16.2

age_group × frailty — 18-44 / non_frail

Decile	n	Observed	Expected	Observed rate	Expected rate	Difference
1	25	0	0.1	0.0000	0.0034	-0.1
2	41	0	0.5	0.0000	0.0122	-0.5
3	34	0	0.8	0.0000	0.0225	-0.8
4	40	0	1.3	0.0000	0.0335	-1.3
5	18	0	0.7	0.0000	0.0400	-0.7
6	34	0	1.6	0.0000	0.0465	-1.6
7	37	0	2.1	0.0000	0.0572	-2.1
8	37	0	2.7	0.0000	0.0738	-2.7
9	33	2	3.6	0.0606	0.1102	-1.6
10	32	0	7.1	0.0000	0.2220	-7.1

age_group × frailty — 45-64 / non_frail

Decile	n	Observed	Expected	Observed rate	Expected rate	Difference
1	45	0	0.1	0.0000	0.0029	-0.1
2	36	0	0.4	0.0000	0.0100	-0.4
3	34	0	0.5	0.0000	0.0150	-0.5
4	48	0	1.1	0.0000	0.0225	-1.1
5	41	0	1.3	0.0000	0.0323	-1.3
6	41	0	1.7	0.0000	0.0426	-1.7
7	37	0	2.1	0.0000	0.0557	-2.1
8	44	0	3.2	0.0000	0.0720	-3.2
9	44	0	4.9	0.0000	0.1115	-4.9
10	40	3	8.7	0.0750	0.2178	-5.7

age_group × frailty — 45-64 / pre_frail

Decile	n	Observed	Expected	Observed rate	Expected rate	Difference
1	152	0	0.9	0.0000	0.0058	-0.9
2	128	1	2.2	0.0078	0.0175	-1.2
3	144	1	3.9	0.0069	0.0273	-2.9
4	105	3	3.9	0.0286	0.0374	-0.9
5	148	0	7.3	0.0000	0.0493	-7.3
6	139	0	9.4	0.0000	0.0678	-9.4
7	111	3	9.7	0.0270	0.0869	-6.7
8	144	5	16.6	0.0347	0.1150	-11.6
9	139	1	23.8	0.0072	0.1715	-22.8
10	128	19	38.9	0.1484	0.3038	-19.9

age_group × frailty — 65-79 / pre_frail

Decile	n	Observed	Expected	Observed rate	Expected rate	Difference
1	173	1	1.1	0.0058	0.0064	-0.1
2	83	0	1.2	0.0000	0.0150	-1.2
3	208	0	5.2	0.0000	0.0252	-5.2
4	130	1	4.9	0.0077	0.0378	-3.9
5	134	1	6.7	0.0075	0.0499	-5.7
6	126	2	8.1	0.0159	0.0646	-6.1
7	152	0	12.8	0.0000	0.0841	-12.8
8	142	4	16.6	0.0282	0.1170	-12.6
9	146	11	25.2	0.0753	0.1727	-14.2
10	140	24	46.8	0.1714	0.3340	-22.8

age_group × frailty — 65-79 / frail

Decile	n	Observed	Expected	Observed rate	Expected rate	Difference
1	322	1	4.2	0.0031	0.0131	-3.2
2	243	3	6.7	0.0123	0.0277	-3.7
3	343	4	13.6	0.0117	0.0396	-9.6
4	283	3	15.6	0.0106	0.0552	-12.6
5	318	3	22.9	0.0094	0.0720	-19.9
6	289	3	26.4	0.0104	0.0912	-23.4
7	323	9	38.3	0.0279	0.1185	-29.3
8	305	14	49.3	0.0459	0.1618	-35.3
9	296	35	67.6	0.1182	0.2285	-32.6
10	310	129	128.5	0.4161	0.4145	+0.5

age_group × frailty — 80+ / frail

Decile	n	Observed	Expected	Observed rate	Expected rate	Difference
1	65	0	1.3	0.0000	0.0204	-1.3
2	60	2	2.4	0.0333	0.0405	-0.4
3	72	2	4.0	0.0278	0.0551	-2.0
4	50	0	3.5	0.0000	0.0698	-3.5
5	71	1	6.4	0.0141	0.0902	-5.4
6	57	0	6.5	0.0000	0.1146	-6.5
7	70	4	10.8	0.0571	0.1538	-6.8
8	61	10	12.9	0.1639	0.2109	-2.9
9	61	14	17.4	0.2295	0.2848	-3.4
10	62	12	27.2	0.1935	0.4394	-15.2

age_group × frailty — 65-79 / non_frail

Decile	n	Observed	Expected	Observed rate	Expected rate	Difference
1	23	0	0.1	0.0000	0.0028	-0.1
2	27	0	0.3	0.0000	0.0126	-0.3
3	16	0	0.3	0.0000	0.0200	-0.3
4	24	0	0.7	0.0000	0.0277	-0.7
5	19	0	0.7	0.0000	0.0374	-0.7
6	27	1	1.3	0.0370	0.0476	-0.3
7	18	0	1.1	0.0000	0.0594	-1.1
8	21	0	1.8	0.0000	0.0855	-1.8
9	23	0	2.7	0.0000	0.1191	-2.7
10	20	0	4.0	0.0000	0.2022	-4.0

age_group × frailty — 80+ / non_frail

Decile	n	Observed	Expected	Observed rate	Expected rate	Difference
1	3	0	0.0	0.0000	0.0083	-0.0
2	1	0	0.0	0.0000	0.0150	-0.0
3	2	0	0.0	0.0000	0.0200	-0.0
4	2	0	0.1	0.0000	0.0250	-0.1
5	3	0	0.2	0.0000	0.0633	-0.2
6	1	0	0.1	0.0000	0.0750	-0.1
7	2	0	0.2	0.0000	0.1050	-0.2
8	2	0	0.3	0.0000	0.1550	-0.3
9	1	0	0.2	0.0000	0.2350	-0.2
10	1	0	0.3	0.0000	0.2600	-0.3

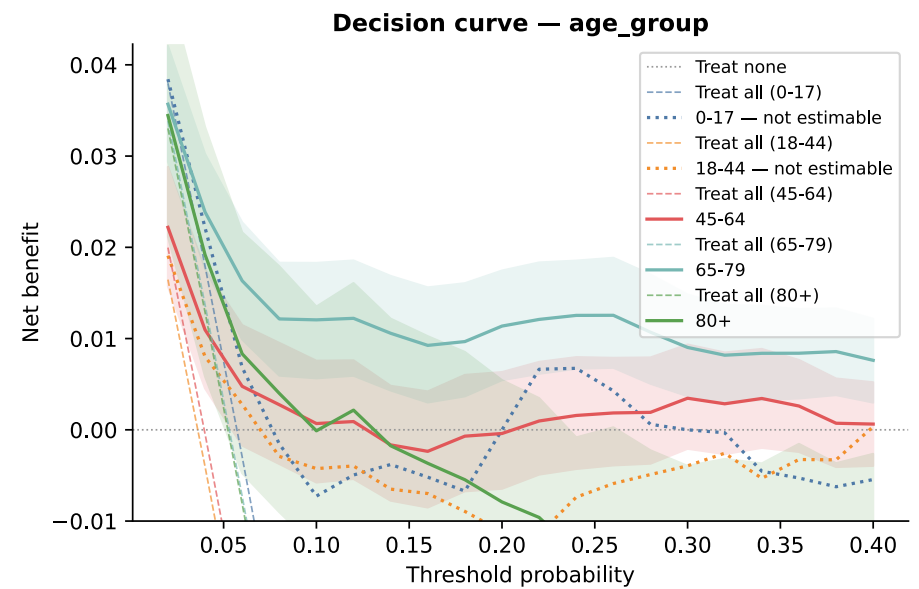
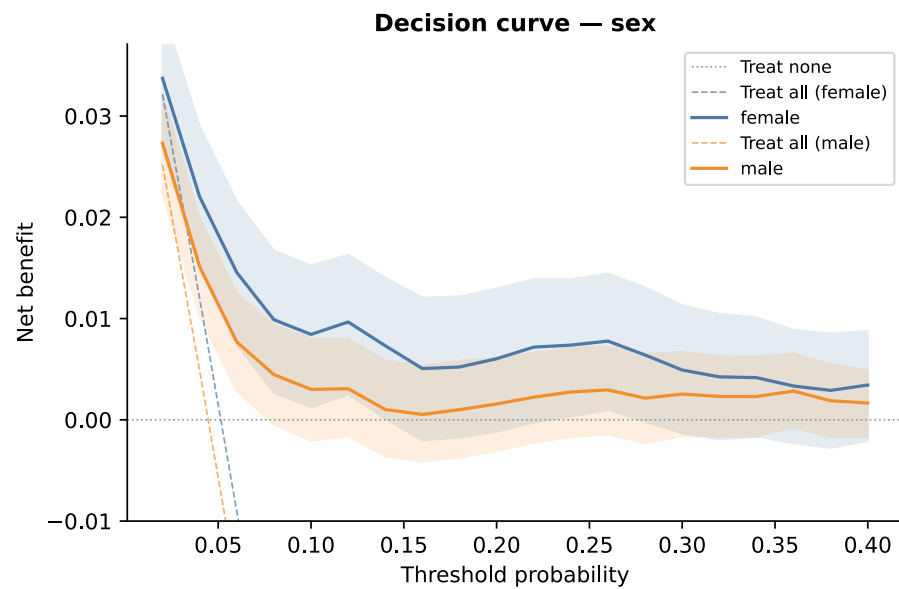
age_group × frailty — 80+ / pre_frail

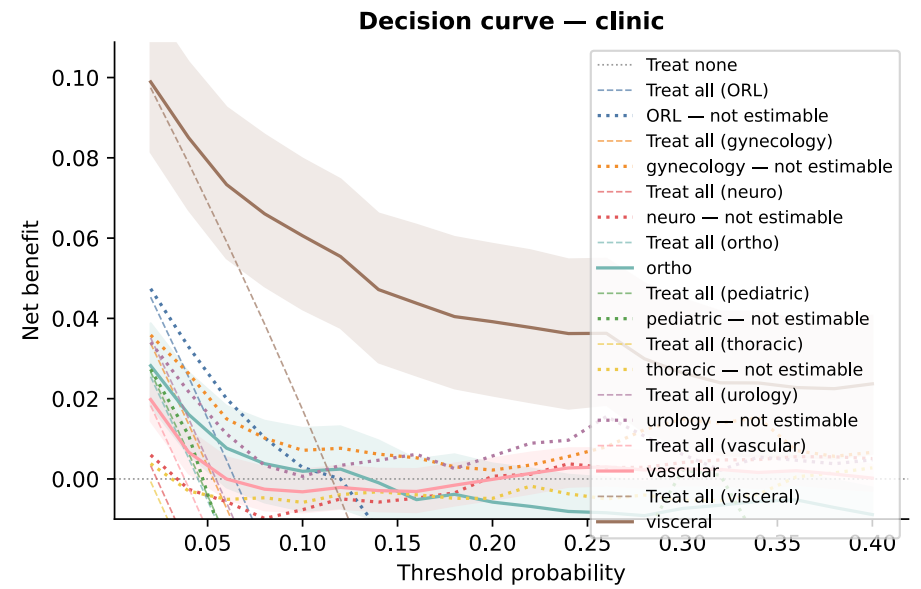
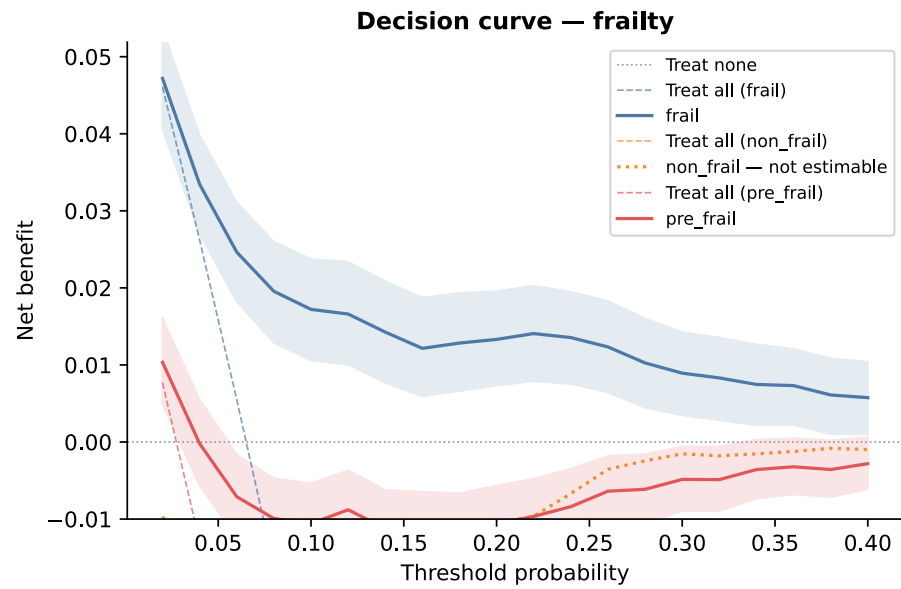
Decile	n	Observed	Expected	Observed rate	Expected rate	Difference
1	42	0	0.4	0.0000	0.0100	-0.4
2	34	1	0.8	0.0294	0.0229	+0.2
3	39	0	1.3	0.0000	0.0323	-1.3
4	38	0	1.7	0.0000	0.0455	-1.7
5	28	1	1.6	0.0357	0.0579	-0.6
6	35	0	2.5	0.0000	0.0711	-2.5
7	41	1	3.8	0.0244	0.0924	-2.8
8	36	0	4.5	0.0000	0.1257	-4.5
9	37	2	7.1	0.0541	0.1909	-5.1
10	35	3	10.9	0.0857	0.3113	-7.9

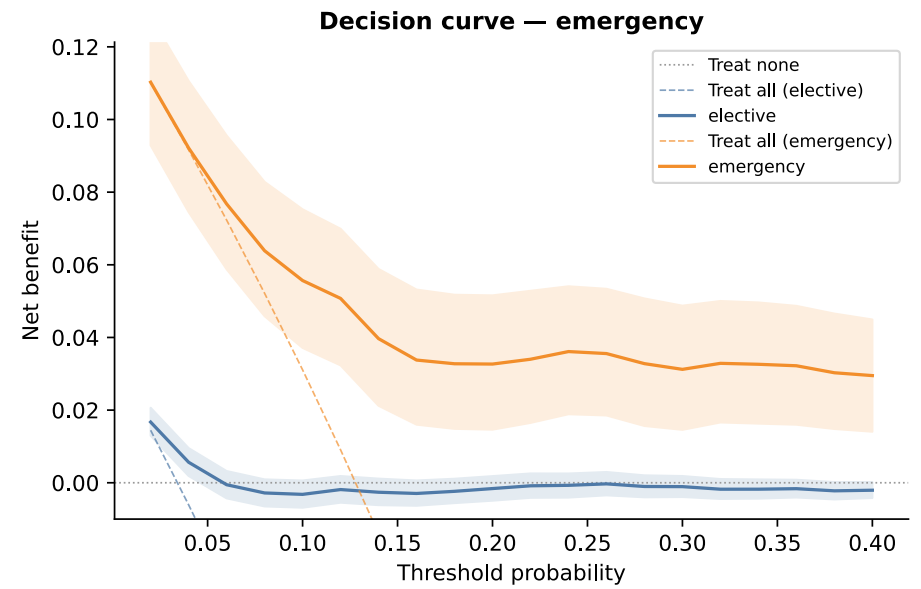
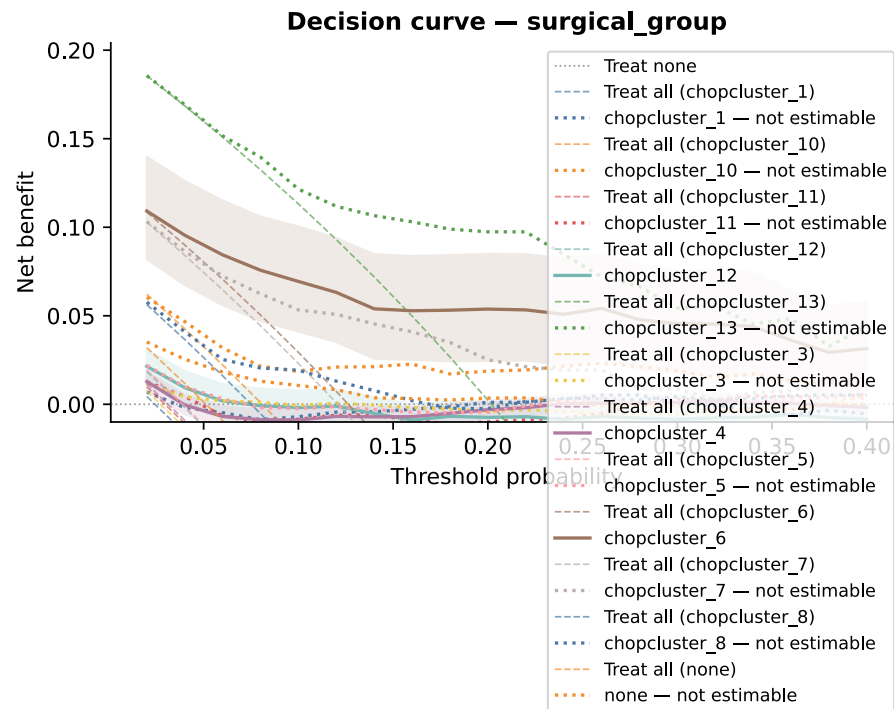
Clinical utility (decision curves)

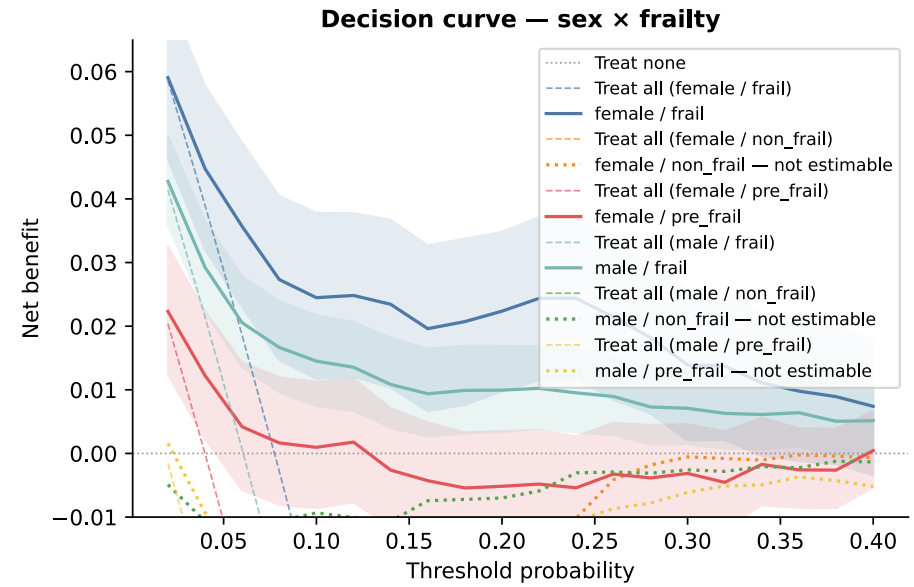
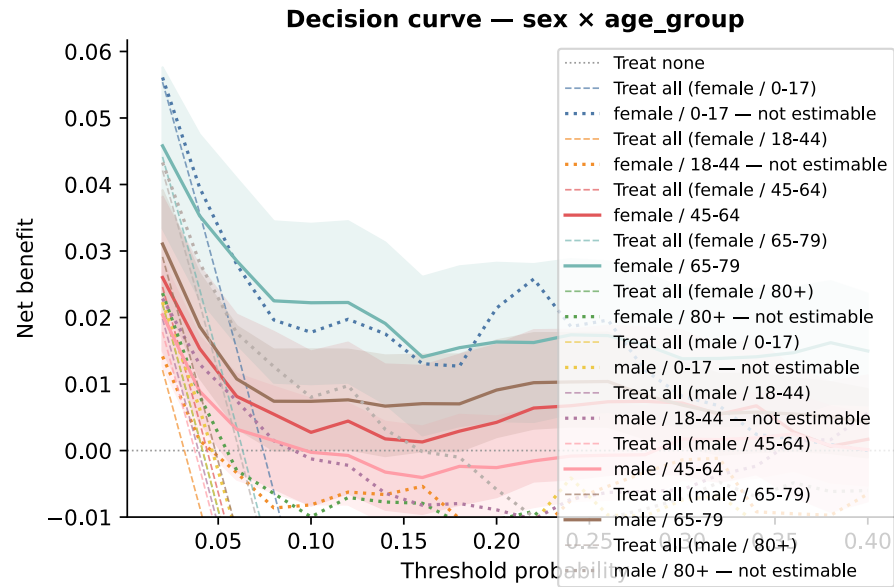
Clinical utility gap statistics

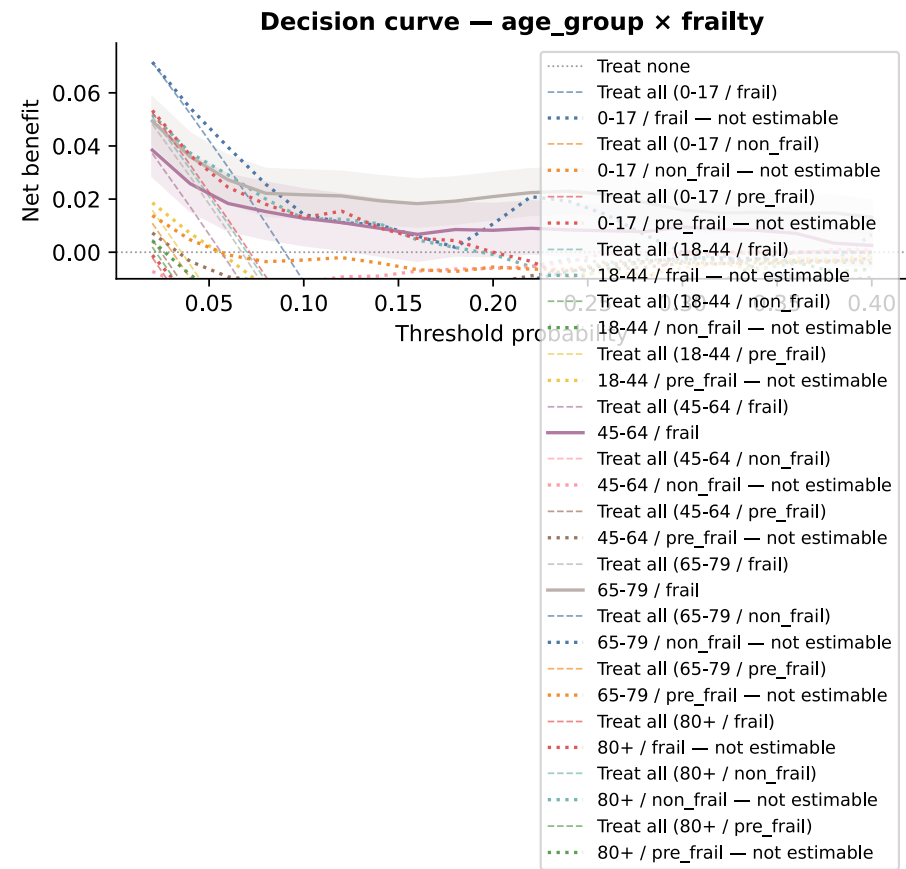
Attribute	Metric	Value (95% CI)
sex	max_net_benefit_gap	0.0069 (0.0030–0.0164)
age_group	max_net_benefit_gap	0.0260
frailty	max_net_benefit_gap	0.0569
clinic	max_net_benefit_gap	0.0951
surgical_group	max_net_benefit_gap	0.1787
emergency	max_net_benefit_gap	0.0936 (0.0761–0.1116)
sex × age_group	max_net_benefit_gap	0.0523
sex × frailty	max_net_benefit_gap	0.0733
age_group × frailty	max_net_benefit_gap	0.0911











Interaction cell diagnostics

Attribute	Subgroup	n	Events	Shrunk	Marginal rate	Cell rate	Shrinkage factor
sex × age_group	female / 0-17	175	13	Yes	0.0515	0.0743	0.5020
sex × age_group	female / 18-44	408	13	Yes	0.0515	0.0319	0.3018
sex × age_group	female / 45-64	1179	52	No	0.0515	0.0441	0.1301
sex × age_group	female / 65-79	1470	93	No	0.0515	0.0633	0.1071
sex × age_group	female / 80+	457	19	Yes	0.0515	0.0416	0.2785
sex × age_group	male / 0-17	192	8	Yes	0.0447	0.0417	0.4788
sex × age_group	male / 18-44	533	21	Yes	0.0447	0.0394	0.2487
sex × age_group	male / 45-64	2536	95	No	0.0447	0.0375	0.0650
sex × age_group	male / 65-79	3214	156	No	0.0447	0.0485	0.0520
sex × age_group	male / 80+	555	34	Yes	0.0447	0.0613	0.2412
sex × frailty	female / frail	1666	129	No	0.0515	0.0774	0.1119
sex × frailty	female / non_frail	522	1	Yes	0.0515	0.0019	0.2869
sex × frailty	female / pre_frail	1501	60	No	0.0515	0.0400	0.1227
sex × frailty	male / frail	4473	271	No	0.0447	0.0606	0.0448
sex × frailty	male / non_frail	500	5	Yes	0.0447	0.0100	0.2958
sex × frailty	male / pre_frail	2057	38	Yes	0.0447	0.0185	0.0926
age_group × frailty	0-17 / frail	211	19	Yes	0.0572	0.0900	0.3891
age_group × frailty	0-17 / non_frail	45	0	Yes	0.0572	0.0000	0.7492
age_group × frailty	0-17 / pre_frail	111	2	Yes	0.0572	0.0180	0.5477
age_group × frailty	18-44 / frail	300	21	Yes	0.0361	0.0700	0.3094
age_group × frailty	18-44 / non_frail	331	2	Yes	0.0361	0.0060	0.2888
age_group × frailty	18-44 / pre_frail	310	11	Yes	0.0361	0.0355	0.3024
age_group × frailty	45-64 / frail	1967	111	No	0.0396	0.0564	0.0640
age_group × frailty	45-64 / non_frail	410	3	Yes	0.0396	0.0073	0.2469
age_group × frailty	45-64 / pre_frail	1338	33	Yes	0.0396	0.0247	0.0913
age_group × frailty	65-79 / frail	3032	204	No	0.0532	0.0673	0.0424
age_group × frailty	65-79 / non_frail	218	1	Yes	0.0532	0.0046	0.3814
age_group × frailty	65-79 / pre_frail	1434	44	Yes	0.0532	0.0307	0.0857
age_group × frailty	80+ / frail	629	45	Yes	0.0524	0.0715	0.1761
age_group × frailty	80+ / non_frail	18	0	Yes	0.0524	0.0000	0.8819
age_group × frailty	80+ / pre_frail	365	8	Yes	0.0524	0.0219	0.2691

Appendix: TRIPOD+AI item coverage

Fairness-relevant items from the TRIPOD+AI checklist (Collins et al., 2024, *BMJ*). Items addressed by this audit are linked to the relevant report section. Items marked "User responsibility" require information that the audit tool cannot produce automatically.

Item	Description	Status	Detail
3c	Describe any known health inequalities between sociodemographic groups	<i>User responsibility</i>	User must provide context in manuscript introduction
5a	Describe data sources, rationale, and representativeness	<i>User responsibility</i>	User must describe data representativeness in manuscript methods
7	Describe data pre-processing, including whether similar across sociodemographic groups	<i>User responsibility</i>	User must verify preprocessing does not vary by group
8a	Outcome assessment consistency across sociodemographic groups	<i>User responsibility</i>	User must verify outcome labels are measured consistently
8b	Assessor demographics for outcome assessment	<i>User responsibility</i>	User must report assessor demographics
9c	Assessor demographics for predictor measurement	<i>User responsibility</i>	User must report assessor demographics
12e	All measures used to evaluate model performance	Addressed	Discrimination, Calibration, and Clinical utility sections
12f	Describe any model updating (eg recalibration) for sociodemographic groups or settings	Not included	No mitigation requested
14	Fairness approaches and rationale	Addressed	Audit summary and gap statistics throughout
20b	Report demographic characteristics and group differences	Addressed	Discrimination section (subgroup sizes, prevalences)
20c	Compare predictor distributions including demographics	<i>User responsibility</i>	User must report predictor distributions per group in manuscript
23a	Report performance estimates with CIs, including for key sociodemographic subgroups	Addressed	Discrimination, Calibration, and Clinical utility sections
25	Overall interpretation including fairness issues	<i>User responsibility</i>	User must interpret results in manuscript discussion
26	Limitations including non-representative sample, bias effects	<i>User responsibility</i>	User must discuss limitations in manuscript

Source: Collins GS, Moons KGM, Dhiman P, et al. TRIPOD+AI statement: updated guidance for reporting clinical prediction models that use regression or machine learning methods. *BMJ* 2024;385:e078378. doi: [10.1136/bmj-2023-078378](https://doi.org/10.1136/bmj-2023-078378). This checklist covers fairness-relevant items only; see the full statement for the complete checklist.

Appendix: How to read this report

Discrimination

Discrimination measures how well the model separates positive from negative cases.

AUROC (Area Under the ROC Curve)

Probability that the model ranks a random positive case higher than a random negative case. Range [0, 1]; 0.5 = no discrimination; 1.0 = perfect separation.

AUPRC (Area Under the Precision-Recall Curve)

Summarises precision across all recall levels. More informative than AUROC for low-prevalence outcomes. Higher is better; baseline equals the prevalence.

Sensitivity (True Positive Rate)

Proportion of true positive cases correctly identified at the chosen threshold. = $TP / (TP + FN)$.

Specificity (True Negative Rate)

Proportion of true negative cases correctly identified at the chosen threshold. = $TN / (TN + FP)$.

PPV (Positive Predictive Value)

Proportion of positive predictions that are truly positive. = $TP / (TP + FP)$.

NPV (Negative Predictive Value)

Proportion of negative predictions that are truly negative. = $TN / (TN + FN)$.

Calibration

Calibration measures whether predicted probabilities match observed event rates.

Calibration intercept

Intercept from logistic recalibration with slope fixed to 1 (Steyerberg 2009). Perfect calibration = 0. Positive values indicate the model underestimates risk on average; negative values indicate overestimation.

Calibration slope

Slope from logistic recalibration with free intercept and slope. Perfect calibration = 1. Values < 1 indicate overconfident predictions (predicted probabilities too extreme); values > 1 indicate underconfident predictions.

Brier score

Mean squared difference between predicted probabilities and binary outcomes. Range [0, 1]; lower is better.

ECE (Expected Calibration Error)

Weighted average of absolute calibration error across equal-frequency bins. Lower is better.

ICI (Integrated Calibration Index)

Mean absolute difference between LOESS-smoothed observed probabilities and predicted probabilities (Austin & Steyerberg 2019). Lower is better; more robust than ECE as it avoids binning artefacts.

Hosmer–Lemeshow statistic

Chi-square-distributed goodness-of-fit statistic comparing observed with expected events across deciles of predicted risk. A large value indicates lack of fit. **It scales with sample size**, so it is not comparable between subgroups of different size and must be read alongside n , the per-decile table, and the calibration measures above (Kramer & Zimmerman 2007).

Hosmer–Lemeshow degrees of freedom

Equal to the number of risk groups used. The classical convention subtracts 2, which is the penalty for estimating an intercept and slope on the data under test; this package evaluates predictions supplied by a model fitted elsewhere, so no degrees of freedom are spent and none are subtracted.

Decile slope and intercept

Least-squares regression of the observed on the expected event rate across risk groups, recommended by Kramer & Zimmerman (2007) as an adjunct to the statistic. Perfect calibration gives a slope of 1 and an intercept of 0.

Clinical utility (decision curves)

Decision curve analysis quantifies the clinical value of using the model at a given decision threshold.

Net benefit

$TP/n - FP/n \times p_t / (1 - p_t)$, where p_t is the decision threshold. Higher is better. A model is clinically useful when its net benefit exceeds both 0 (treat none) and the treat-all reference.

Standardised net benefit

Net benefit divided by prevalence (Naderalvojud 2025). Normalised to a [0, 1] scale for comparison across cohorts with different prevalences.

Treat-all reference

Net benefit if all patients receive treatment. = prevalence – (1 – prevalence) $\times p_t / (1 - p_t)$. Becomes strongly negative at high thresholds.

Treat-none reference

Net benefit = 0. The baseline of no intervention.

Gap statistics

Gap statistics quantify the largest disparity between subgroups for each metric. All gaps are non-negative; 0 = perfect parity.

Equal opportunity difference

max – min of sensitivity across subgroups. Measures fairness in true positive identification.

Predictive equality

max – min of false positive rate (1 – specificity) across subgroups. Measures fairness in false alarm rates.

Equalized odds gap

Maximum of equal opportunity difference and predictive equality.

Statistical parity

max – min of positive prediction rate across subgroups. Measures whether the model predicts positive at equal rates.

Max intercept / slope gap

Maximum pairwise difference in calibration intercept or slope across subgroups, with bootstrap confidence intervals.

Max net benefit gap

Maximum across all thresholds of the maximum pairwise net-benefit difference between subgroups, with bootstrap confidence intervals.

Confidence intervals and shrinkage**Bootstrap confidence intervals**

95% percentile intervals from subgroup-stratified patient-level resamples. Patients are resampled with replacement within each subgroup, preserving the subgroup's size while letting its event count vary.

Empirical Bayes shrinkage

Interaction cells with fewer events than the configured threshold (shown in *italics*) use Beta-Binomial shrinkage toward the marginal subgroup rate. This reduces instability in small cells. Confidence intervals are suppressed for shrunk estimates.

Generated by **isitfair**. Three-axis evaluation: discrimination, calibration, and clinical utility reported jointly per TRIPOD+AI recommendations.

Bootstrap confidence intervals (95%) from 2000 subgroup-stratified patient-level resamples. Interaction cells with fewer than the configured event threshold use empirical Bayes shrinkage (italicised rows).